

Loans4India — Platform Flow v4 (Source of Truth)

Oct 3, 2026 · @JP

How to read this document

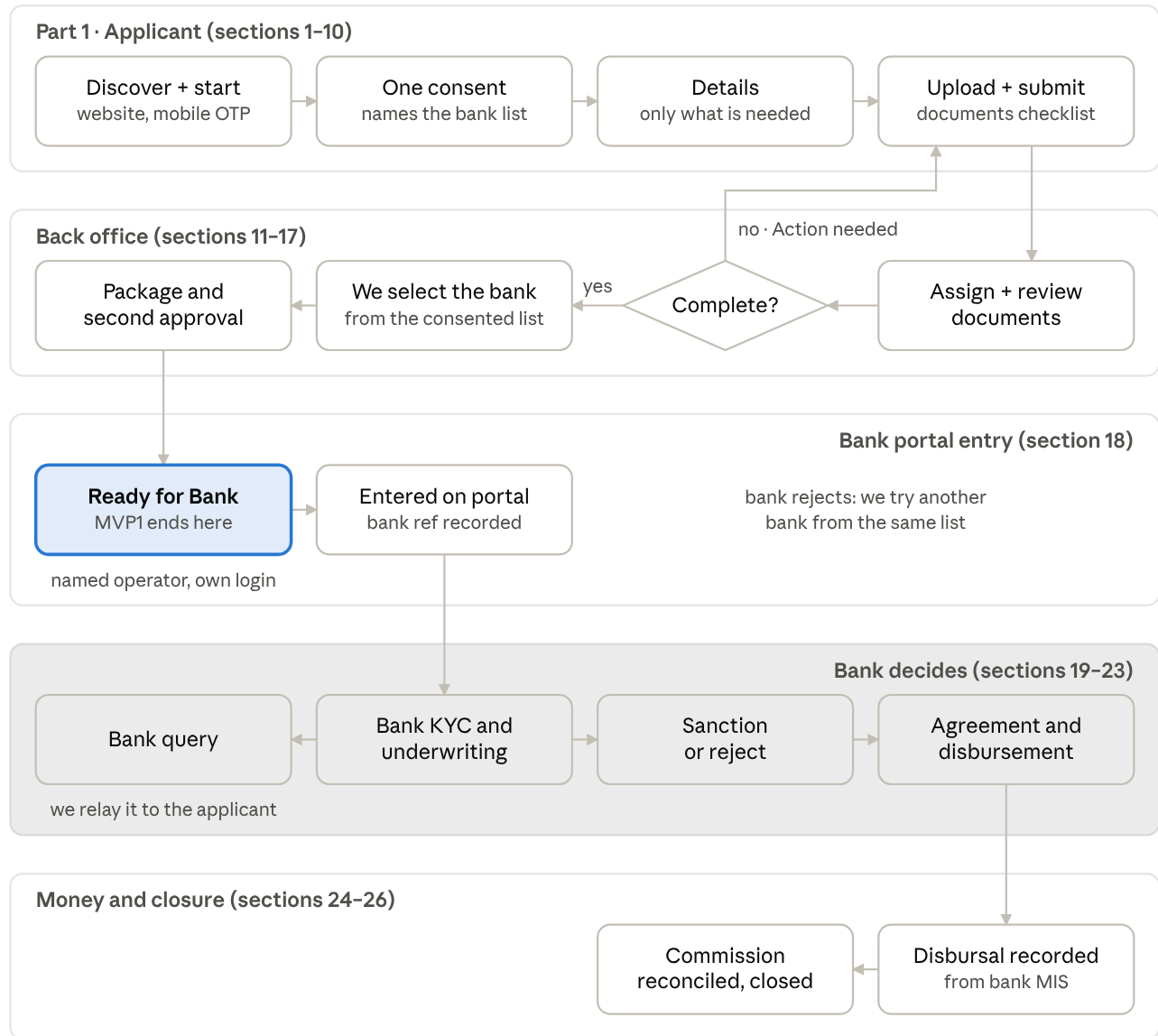
This is the single source of truth for how Loans4India works, written for business, operations and the development team. It assumes no prior knowledge of lending: the glossary below explains every term, every flow has a diagram, and every feature says what it does and what it connects to. It replaces [Loans4India — Platform Flow v3.2](#) . Technical design lives in [Loans4India — System Design Document \(v3.2\)](#) , and the full permission matrix in [Loans4India — RBAC & PAM Spec \(MVP1\)](#) .

The model in one paragraph. Loans4India is a Virtual Corporate DSA: a loan sourcing company working under contract with banks and NBFCs. The applicant gives consent and uploads documents. Loans4India checks the file, picks a partner bank from the list the applicant agreed to, and keys the application into that bank's portal. The bank does its own checks and makes every decision. When the bank disburses the loan, it pays Loans4India a commission. Loans4India never lends, never decides, and never charges the applicant.

Each section follows the same pattern: *In plain words* (what and why) → diagram → *Step by step* → *Rules* → *Feature map* (what each feature does and what it links to) → *Edge cases*, where relevant. *Proposed default* marks a number to confirm; **VERIFY** marks a legal point for counsel.

The platform at a glance

The applicant uploads; Loans4India does the rest; the bank decides



the platform at a glance · where each part of this doc sits

What changed from the old outline

Old heading	In v4	Why
3. Quick Eligibility	3. Quick estimate (indicative only)	We don't assess eligibility; the bank does. A simple calculation is allowed if clearly labelled
4. Registration, 5. Customer Profile, 11. Full Application	4. Registration, then 6. Application details (one section)	Profile and full application were the same form twice; the applicant fills it once
6. Consent + Credit Check	5. Consent	No credit bureau pull: Loans4India isn't allowed to and doesn't need to
7. Eligibility Engine	Removed	The bank alone decides eligibility
9. Customer Sees Matched Offers	Removed	No offers are shown; Loans4India picks one bank from the consented list
19. Lender-Ready Gate, 20. Lender Eligibility Check, 21. Lender Assignment	15. Bank selection, 16. Submission package, 17. Ready-for-Bank gate	Reordered to match the real flow: pick the bank, build its package, then approve
23. Lender Submission	18. Bank portal entry	Submission is manual entry on the bank's DSA portal
36. Document/KYC Reviewer	30. Document Reviewer	KYC is the bank's job; we only check quality
(missing)	10, 31, 41-55	Status tracker, Authorised DSA Operator, data rights, communications, governance, security, affiliates, product configuration, empanelment, AI policy, system requirements, legal, compliance, MVP1, roadmap, metrics, decisions

Glossary

Term	Meaning in plain words
Applicant / customer	The person applying for a loan through Loans4India
Bank / NBFC / lender	The regulated institution that actually lends. An NBFC is a non-bank finance company. In this doc "bank" covers both
DSA (Direct Selling Agent)	A company a bank contracts to find and prepare loan applicants. There is no "DSA licence"; it's a separate contract with each bank
Corporate DSA / DSA code	The DSA agreement with a bank, and the ID the bank gives the DSA to track its applications and pay commission
Master DSA / sub-DSA	A large DSA that already has bank contracts. A smaller DSA can work under it while getting its own contracts
LSP (Lending Service Provider)	RBI's term for a company that helps a lender with digital lending. Loans4India may be treated as one (VERIFY)
Bank DSA portal	The bank's own website where an authorised DSA staff member enters applications. It's outside our system
KYC (Know Your Customer)	Identity verification. The bank does KYC; Loans4India only checks document quality
Underwriting / sanction	The bank's credit assessment, and its formal approval with amount, rate and tenure
In-principle approval	A conditional "yes, subject to checks" from a bank. Not a sanction
Disbursement	The bank paying the loan amount to the applicant
MIS file	A spreadsheet the bank sends with the status of applications

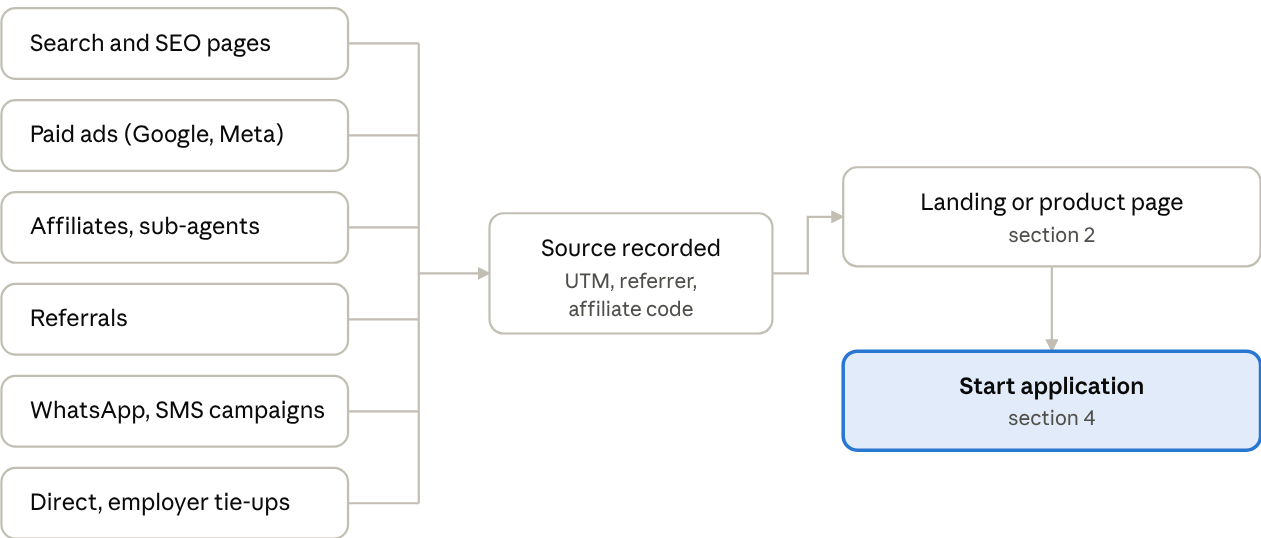
Term	Meaning in plain words
Payout file	A file the bank sends listing the commission it is paying
Commission / clawback	The fee the bank pays Loans4India per disbursed loan, and a recovery of that fee if, say, the loan is cancelled
FOIR	Fixed Obligation to Income Ratio: the share of monthly income already going to EMIs. Used only in the indicative estimate
EMI	Equated Monthly Instalment: the fixed monthly loan repayment
PAN / Aadhaar	India's tax ID and national ID. PAN is stored encrypted; the Aadhaar number is never stored
DPDP	India's Digital Personal Data Protection Act and Rules: consent, purpose limits, data rights
TRAI DLT	The telecom regulator's registry: SMS sender IDs and templates must be registered
CERT-In	India's cyber-security agency; certain incidents must be reported to it
Application state / attempt / bank status	The three status fields: our workflow, the current submission to one bank, and what the bank has confirmed
Action needed	A task asking the applicant to fix or supply something
Purge	Deleting personal data when its retention period ends, keeping only a log that it was done
RBAC / PAM	Role-based access control (who can do what) and privileged access management (tight control of admin powers)
MVP1	The first build: everything up to Ready for Bank, on synthetic test data

Part 1 — Customer journey

1. Customer enters the platform

In plain words: people reach Loans4India from many places: Google, ads, partners, friends, messages. Wherever they come from, they land on the same website and start the same application. The platform remembers where each applicant came from, so we know which channels work and which partner earned a payout.

Every channel lands on the same start, with its source recorded



No bank names before agreements. Affiliates get a referral link only, never applicant data.

entry points · six channels, one start

Entry points

Channel	How it reaches us	What we record	Rules
Search and SEO pages	Google results for loan topics, linked to our product pages and guides	Landing page, search referrer	Content approved; no approval promises
Paid ads (Google, Meta)	Ads linking to a product page	UTM source, medium, campaign	Ad copy uses approved claims only; tracking pixels only with visitor consent

Channel	How it reaches us	What we record	Rules
Affiliates and sub-agents	Their unique referral link or code	Affiliate code, link ID	Affiliate is onboarded and contracted first (section 45); never sees applicant data
Referrals	A link shared by an existing applicant or staff member	Referral code	Referral rewards only if approved and disclosed
WhatsApp and SMS campaigns	Messages to people who opted in	Campaign ID	DLT-registered templates; only to people with marketing consent (VERIFY)
Direct and employer tie-ups	Typing the website address; links from employers we partner with	Direct or partner code	Partner agreement required

Step by step

1. The visitor clicks a link or types the address.
2. The website reads the link's tags (UTM, referrer, affiliate or referral code) and stores them in the browser session, not in the address bar, so they survive page changes.
3. When the visitor starts an application, the tags are saved on the application record.
4. Attribution is used for channel reporting (section 54) and affiliate payouts (section 45).

Feature map

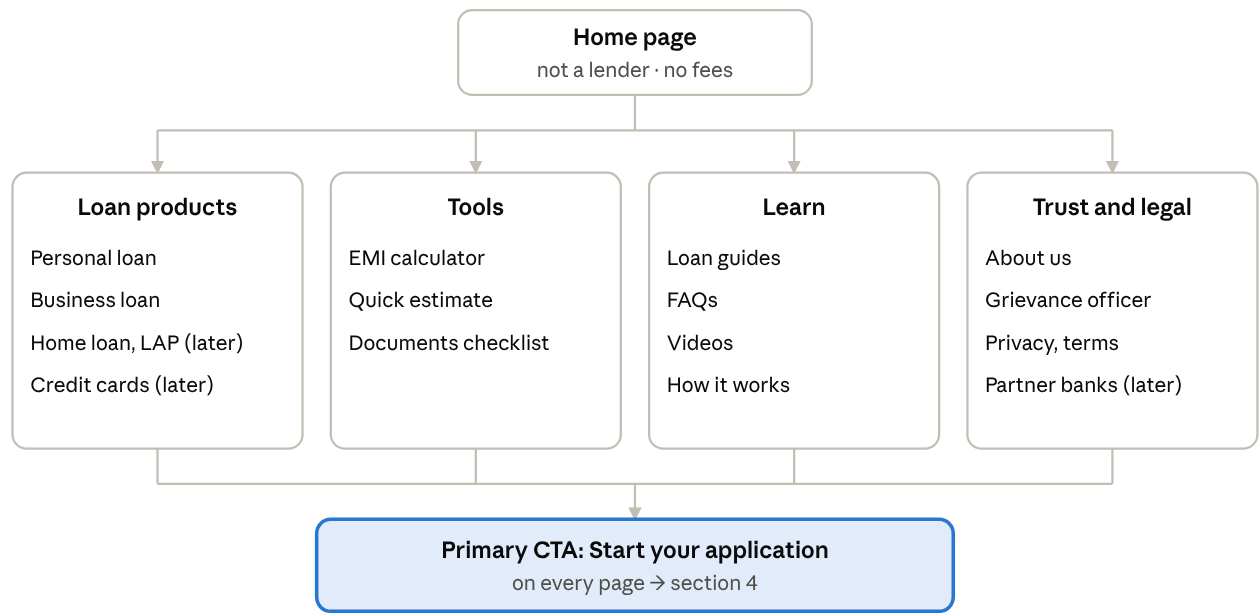
Feature	What it does	Linked to
Attribution capture	Saves source tags on the application	Application record; metrics; affiliate payouts
Referral and affiliate codes	Identify which partner sent the applicant	Affiliate management (section 45)

Feature	What it does	Linked to
Consent-gated analytics	Loads tracking scripts only after the visitor agrees	Data governance (section 43)

2. Customer discovery

In plain words: before applying, people want to understand the loan, check what they might afford, and trust who they’re dealing with. They can browse everything without signing up. Every page has the same main button.

Every page leads to one action: start your application



website map · four page groups, one primary action

Website

Page group	Purpose	Must show	Built today?
Home	Explain the service in one screen	"Loan sourcing service, not a lender"; "No fee from you"; how it works in 3 steps; main button	Yes

Page group	Purpose	Must show	Built today?
Product pages	One page per loan type: who it suits, typical documents, how long it takes	Typical requirements only, never a promised rate or approval	Personal and business loan, yes
EMI calculator	Monthly instalment for an amount, rate and tenure the visitor enters	Labelled as a calculator, not an offer	Yes
Quick estimate	Indicative borrowing range from what the visitor types (section 3)	"Indicative, not an approval"	Yes
Documents checklist	What to keep ready per loan type	Same list the upload screen uses	To build
Guides, FAQs, videos	Answer common questions; SEO	Approved content; FAQ schema for search	FAQ yes
About, grievance, privacy, terms	Legal entity, grievance officer, policies	Real legal entity and officer before launch	Drafts, marked draft
Partner banks	Bank names and logos	Only after agreements and brand-use permission	Hidden (<code>partnersLive: false</code>)

Primary CTA

- **Main button on every page: "Start your application".** It opens section 4.
- **Secondary button: "Estimate what you can borrow",** which opens the quick estimate and then leads to the same start.
- Never "Check eligibility", "Get approved" or "Get instant loan": these imply a credit decision we don't make.
- The header or footer always shows "Loans4India is a loan sourcing service, not a lender. We don't charge you any fee."
- Pages are in English and Hindi, mobile-first, and work at 360 px width.

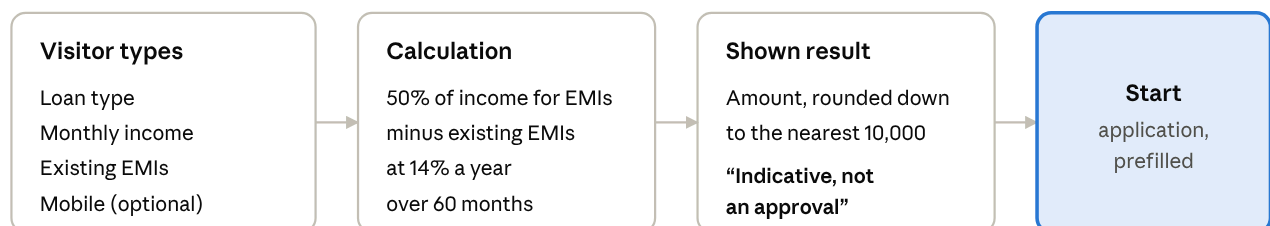
Feature map

Feature	What it does	Linked to
Product pages	Describe each loan type from the product catalogue	Product catalogue (section 46)
EMI calculator	Calculates an instalment in the browser; stores nothing	None (no personal data)
Quick estimate	Indicative range; can prefill the start form	Section 3; section 4
Brand settings (<code>brand.ts</code>)	One place for name, contacts, grievance officer, <code>partnersLive</code> switch	Legal setup (section 50)
SEO (sitemap, robots, FAQ schema)	Helps search engines; blocks indexing of <code>/apply</code> , <code>/admin</code> , <code>/api</code>	Entry points (section 1)

3. Quick estimate (indicative only)

In plain words: a visitor can get a rough idea of how much they might borrow before applying. It's simple arithmetic on what they type, not a check of their credit or a promise from any bank. It replaces the old "Quick Eligibility" step because only the bank can decide eligibility.

The estimate is arithmetic on what the visitor types



Never: a credit bureau pull, a bank name, a rate offer, or storing anything before the visitor starts the application. Values pass to the start form through browser session storage, never in the web address.

quick estimate · inputs, formula, labelled result

How the number is calculated

The affordable monthly instalment is half the monthly income minus existing EMIs. The loan amount is what that instalment repays at the assumed rate and tenure (all *proposed defaults*, held in configuration):

$$\text{EMI}_{\text{max}} = 0.5 \times \text{income} - \text{existing EMIs}$$

$$\text{Amount} = \text{EMI}_{\text{max}} \times \frac{(1+r)^n - 1}{r}, \quad r = \frac{14\%}{12}, \quad n = 60$$

The result is rounded **down** to the nearest ₹10,000. If the instalment comes out at zero or less, the screen says "Your existing EMIs are high; a bank may still consider you. You can still apply."

Rules

- Always labelled "Indicative estimate, not an approval"; never a bank name or a rate offer.
- Runs in the browser. Nothing is saved until the visitor starts the application.
- The mobile field is optional here; if filled, it's passed to the start form with session storage, never in the URL.
- Assumptions (50%, 14%, 60 months) are configuration per product, so they can change without code. **VERIFY** the wording with counsel.

Feature map

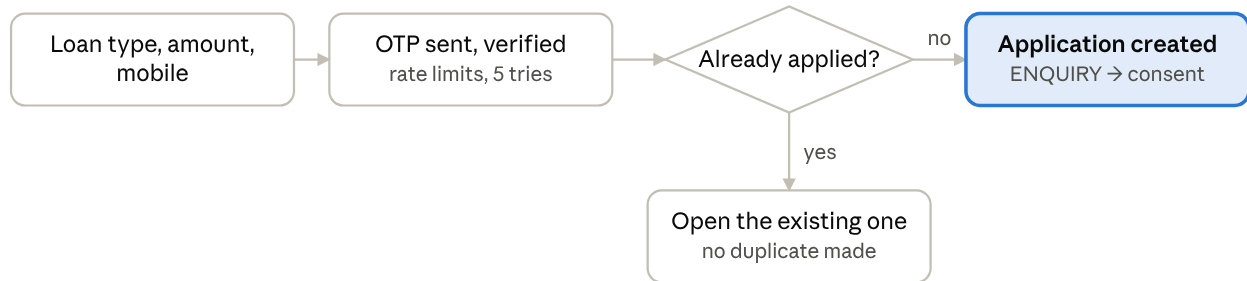
Feature	What it does	Linked to
Estimate widget	Calculates and shows the indicative amount	Product configuration (assumptions); section 2
Prefill handoff	Carries amount, income and mobile to the start form	Section 4
Copy guardrails	Fixed labels and disclaimers	Customer language rules (section 10)

4. Registration (start the application)

In plain words: the applicant picks a loan type, enters the amount and their mobile number, and types the one-time password (OTP) we send. That proves the phone is theirs and creates their application. There's no separate username or password; the same OTP is how they come back later.

A verified mobile opens one application per product

New applicant



Returning applicant



registration and sign-in · OTP for new and returning applicants

Step by step (new applicant)

1. Choose loan type and amount (prefilled if they came from the quick estimate) and enter the mobile number.
2. The platform sends an OTP by SMS through a DLT-registered sender.
3. The applicant enters the OTP. Limits (*proposed defaults*): OTP valid 5 minutes; 5 wrong tries locks that OTP; 3 sends per mobile per 10 minutes; per-IP throttle.
4. If this mobile already has an **open** application for the same product, the applicant is taken to it. No second record is created.
5. Otherwise the platform creates the application with a reference like `LFI-7K3P9Q` (no `0/O/I/I/L`), state `ENQUIRY`, attribution attached, and opens the consent screen (section 5).

Step by step (returning applicant)

1. "Sign in" with the mobile number and OTP.
2. They see only their own applications, each with its status and any open task.
3. Sessions end after 15 minutes idle (*proposed default*).
4. Sensitive actions (data access, correction, erasure, changing the mobile number) need a step-up check: matching PAN or date of birth already on file.

Data captured at this step

Field	Why	Stored as
Loan type, amount	Selects the product checklist	Plain
Mobile	Identity of the account; contact	Encrypted (tier T1), shown masked to staff
Attribution tags	Channel reporting, affiliate payout	Plain
IP, device, time	Abuse protection, consent evidence	Security log

Rules

- Never reveal whether a mobile number already has an account to anyone who hasn't verified the OTP.
- Honeypot field and rate limits against bots (already on the website).
- **Built today:** the website creates the application on form submit without OTP. OTP is a to-do before real applicants.

Feature map

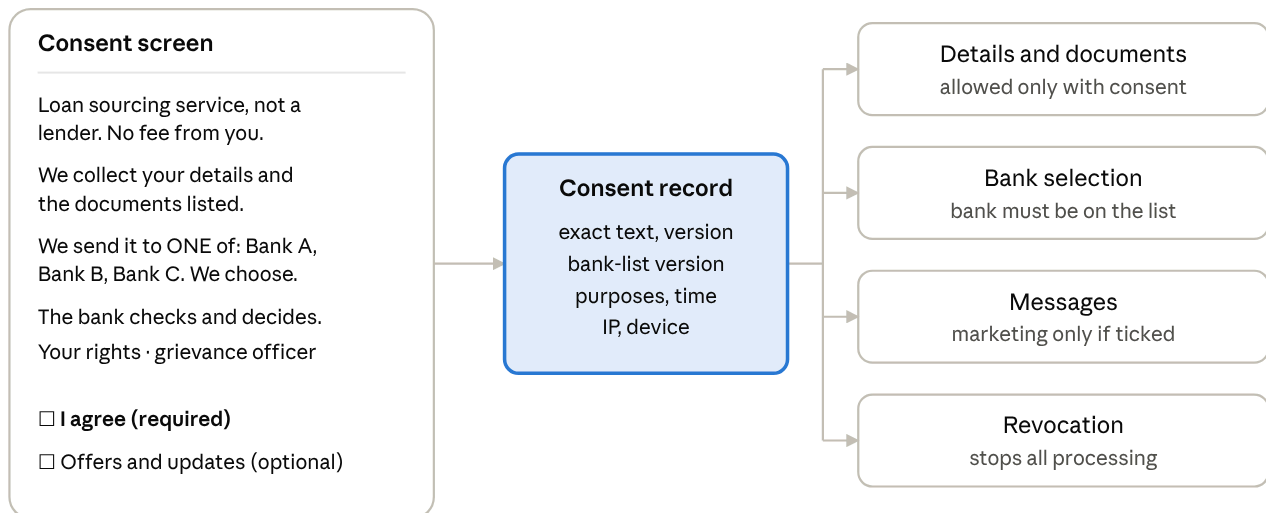
Feature	What it does	Linked to
OTP service	Sends and verifies one-time passwords	SMS provider (DLT); security controls (section 44)

Feature	What it does	Linked to
Duplicate guard	Sends the applicant to their existing open application	Duplicate review in back office (section 11)
Application reference	Human-friendly ID used in all messages	Status tracker; support
Session and step-up	Keeps sign-in short-lived; extra check for sensitive actions	Data rights (section 41)

5. Consent

In plain words: before we collect anything personal, the applicant reads one short screen and ticks one box. It says who we are, what we collect, which banks we may send their application to, and their rights. That tick is the legal basis for everything that follows, so the exact wording and the time it was given are stored permanently. There is **no credit check**: Loans4India never pulls a CIBIL or bureau report.

One consent screen creates one record that every later step checks



consent · one screen, one record, four gates that check it

What the screen must say

1. **Who we are:** a loan sourcing service, not a lender; no fee from the applicant.
2. **What we collect:** the applicant's details and the documents listed for their loan type.

3. **Who receives it:** each partner bank for this product, **named individually** (never just "partner lenders"). We choose one; the bank's name appears in the status after submission. If one bank declines, we may try another on the same list, one at a time.
4. **What the bank does:** its own checks and the decision; the bank may contact the applicant directly.
5. **Rights:** withdraw, revoke consent, access, correct, erase, nominate; how to reach the grievance officer.
6. **Retention:** one line, with the full policy behind a link.
7. **Two ticks:** required consent (application processing, sharing with the listed banks, service messages), and an optional, never pre-ticked marketing consent.

What is stored

Field	Purpose
Exact text shown and its hash	Proves what the applicant agreed to
Notice version, bank-list version	Shows which banks were covered at that moment
Purposes ticked	Separates required from optional (marketing)
Time, IP, device	Evidence of who and when
Status and revoked-at	Whether consent is still valid

Records are **append-only**: never edited, only superseded by a new record or a revocation.

Rules

- Nothing beyond the enquiry (section 4) is processed until the required tick is given. The state moves from `ENQUIRY` to `PROFILE_IN_PROGRESS`.
- A bank added to the list later is **not** covered by older consents; those applicants get a one-tap re-consent request (state `CONSENT_PENDING`) before that bank can be used.
- Revoking consent stops processing at once and starts the exit flow (section 26).
- **Built today:** the website stores 4 purposes (process application, share with partner lenders, contact, optional marketing) with text and version `2026-10-02.v1`. To change: name the banks individually and add the bank-list version.

- **VERIFY:** that one consent naming a bank list, without a notice before each submission, meets DPDP and lender-disclosure rules (fallback: automatic SMS naming the bank at submission).

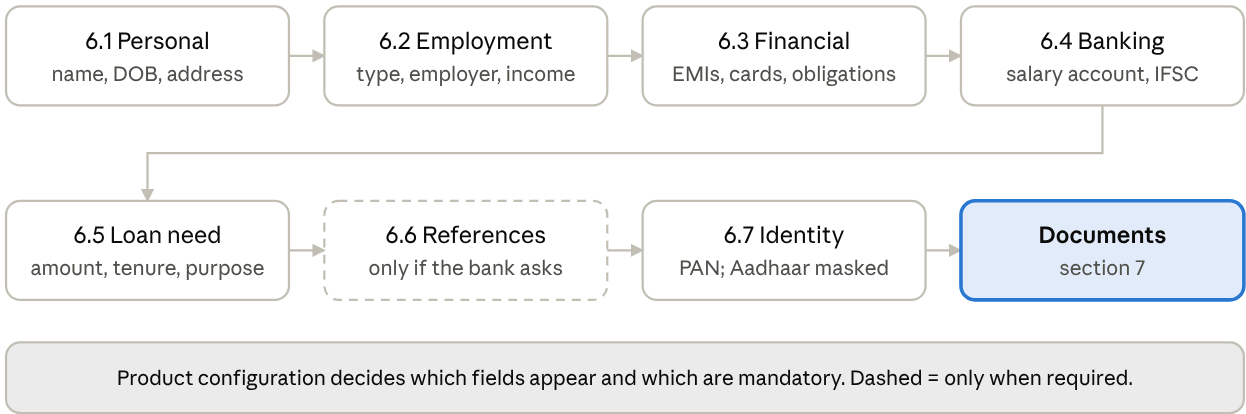
Feature map

Feature	What it does	Linked to
Consent screen	Shows the notice and captures the ticks	Partner-bank list per product (section 46)
Consent record	Stores evidence, append-only	Single source of truth (section 39); audit
Consent check	Every service and background job checks validity first	All later sections
Re-consent request	One tap to approve a newly listed bank	Bank selection (section 15)
Revocation	Withdraws consent from My Data or support	Exits and purge (section 26); data rights (section 41)

6. Application details

In plain words: this is the one form the applicant fills in. It replaces the old "Customer Profile" and "Full Application", which asked for the same things twice. It's split into short sections, saves after every field, and shows only what the chosen product and bank template need. A salaried applicant never sees business questions.

Seven short sections, shown only as the product needs, autosaved



application details · seven sections in order

How the form behaves

- **Autosave:** each field saves when the applicant leaves it, with a version history (who, when, old value, new value).
- **Validation:** formats are checked on screen and again on the server (PAN pattern, PIN code, IFSC, date of birth gives age 21–60, *proposed default*).
- **Mandatory vs optional:** marked on screen; the product configuration decides which.
- **Leave and resume:** the applicant can stop at any point and sign in later (section 4).
- **Tiers:** every field has a sensitivity tier (T0–T3, section 44) that decides masking for staff.

6.1 Personal

Field	Required	Why it's needed	Tier
Full name (as on PAN)	Yes	Matches documents and bank records	T1
Date of birth	Yes	Bank age criteria; PAN match	T2
Gender	Only if the bank template asks	Some bank forms require it	T2

Field	Required	Why it's needed	Tier
Current address, PIN, city	Yes	Bank serviceability by location	T2
Residence type (owned, rented, family, company)	Yes	Common bank form field	T2
Years at current address	If template asks	Stability check by bank	T2
Email	Yes	Messages and bank contact	T1

6.2 Employment

Field	Required	Why	Tier
Employment type (salaried, self-employed professional, self-employed business)	Yes	Decides the rest of the form and documents	T0
Employer or business name	Yes	Bank checks employer category	T2
Designation / nature of business	Yes	Bank form	T2
Industry	If template asks	Bank risk category	T2
Time in current job / business vintage	Yes	Bank minimums	T2
Monthly net income (salaried) or annual income (self-employed)	Yes	Repayment capacity	T2
Salary mode (bank transfer, cash, cheque)	Salaried	Many banks need salary by bank transfer	T2
GST registration (business)	Business loans	Bank requirement	T2

6.3 Financial

Field	Required	Why	Tier
Existing loans (type, lender, EMI)	Yes ("none" allowed)	Obligations the bank will check	T2
Total existing EMIs per month	Yes	Shown to the bank; matches the estimate	T2
Credit cards held (count, total limit)	If template asks	Bank form	T2
Other regular obligations	Optional	Bank form	T2

6.4 Banking

Field	Required	Why	Tier
Salary / primary bank name	Yes	Matches bank statements	T2
Account number	If template asks	Some banks need it for disbursal set-up	T3
IFSC	If account number asked	Validates the branch	T2

6.5 Loan requirement

Field	Required	Why	Tier
Amount	Yes	From registration; editable	T0
Preferred tenure	Yes	Bank form	T0
Purpose (from a list)	Yes	Bank form; some purposes not financed	T0

6.6 References (only where the bank requires)

Field	Required	Why	Tier
Two references: name, relationship, mobile, city	Only if the bank template lists them	Bank form	T1 (reference's own data)

The applicant is told references may be contacted by the bank; only the minimum is collected.

6.7 Identity details (KYC identifiers)

Field	Required	Stored as	Notes
PAN	Yes	Encrypted, plus a keyed hash for duplicate checks	Checked against name through a licensed provider (section 14)
Aadhaar	Only if the bank template needs it, as a masked document	Never the full number ; masked image only	No Aadhaar authentication by Loans4India
Father's / spouse's name	If template asks	T2	Bank forms

This is **not** KYC. It collects the identifiers the bank needs; the bank does KYC itself.

Feature map

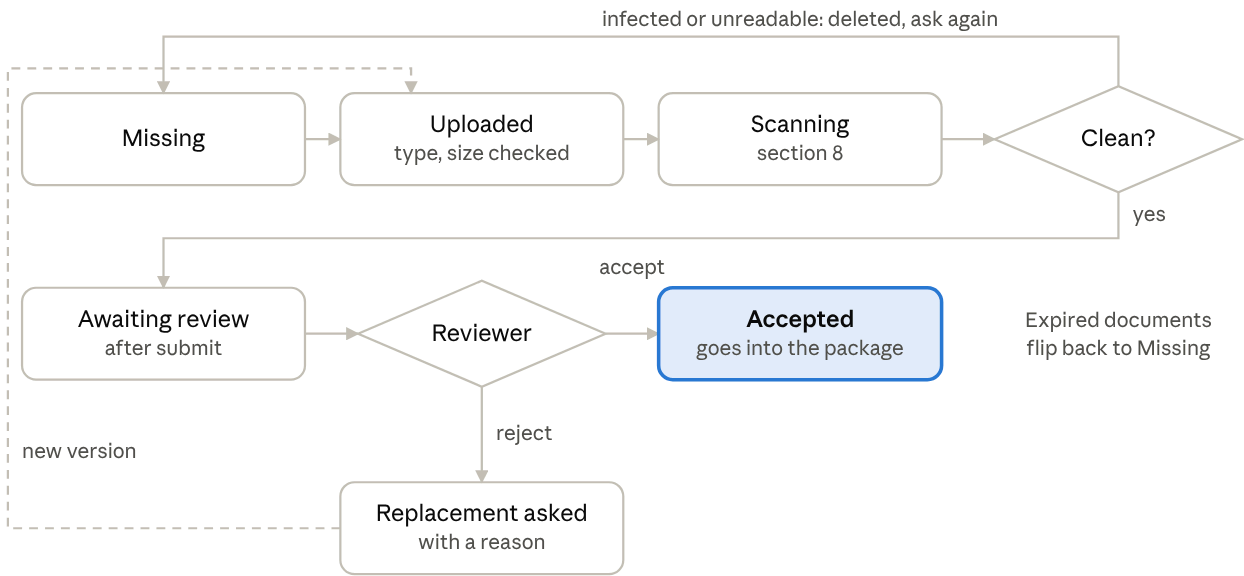
Feature	What it does	Linked to
Dynamic form	Shows fields from the product and bank template	Product configuration (section 46)
Autosave and field history	Saves each field with versions	Single source of truth (section 39); audit
Server validation	Rejects bad formats even if the browser is bypassed	Security (section 44)

Feature	What it does	Linked to
PAN encryption and hash	Protects PAN; allows duplicate matching without reading it	Duplicate guard (sections 4, 11)
Masking by tier	Controls what staff see	RBAC (section 27)

7. Document center

In plain words: the applicant sees a checklist of documents for their loan type, one "slot" per document. Each slot shows its status: missing, uploading, being checked, accepted, or needs replacing. Documents are stored privately and never shared by link.

Each document slot moves from missing to accepted



document slot lifecycle · missing to accepted

Typical checklist (salaried personal loan; the bank template decides the final list)

Slot	Typical requirement	Expiry rule (<i>proposed default</i>)
PAN card	Clear image or PDF, both sides if applicable	Doesn't expire

Slot	Typical requirement	Expiry rule (<i>proposed default</i>)
Address proof	Passport, driving licence, voter ID, utility bill, or masked Aadhaar	Utility bill under 3 months old
Photograph	Recent, if the bank asks	—
Salary slips	Last 3 months	Latest slip under 45 days old
Bank statements	Last 6 months of the salary account	Latest entry under 30 days old
Form 16 / ITR	If the bank template asks	Latest financial year
Business documents (business loans)	GST returns, ITR, registration certificate	Per bank

Step by step

1. The applicant opens a slot and takes a photo or picks a file (PDF, JPG, PNG; up to 10 MB; several pages allowed).
2. The file uploads straight to a private quarantine area through a short-lived link (section 8).
3. The slot shows "Checking" while it's scanned. If it's infected, corrupt or unreadable, it's deleted and the applicant is asked to upload again.
4. Clean files show as "Uploaded". The applicant can replace a file before submitting; old versions are kept.
5. After submission, a reviewer accepts the file or asks for a replacement with a reason (section 14).
6. If a document expires before submission to the bank, its slot flips back to Missing and the applicant is told.

Rules

- **Password-protected PDFs** (common for bank statements): the applicant enters the password once; the file is unlocked on the server and the password is never stored.
- Aadhaar is accepted only as a masked copy (first 8 digits hidden); the full number is never stored.

- Only the latest accepted version of each slot goes into a bank package. Rejected versions stay in history but can't be used.
- Staff see documents in a watermarked preview; downloads need approval (section 44).
- **VERIFY:** how long documents may be kept if Loans4India is treated as an LSP (section 43).

Feature map

Feature	What it does	Linked to
Checklist	Lists slots from the product and bank template	Product configuration (section 46)
Direct upload	Sends the file to quarantine without passing through our servers	Security pipeline (section 8)
Slot status	Shows progress per document	Status tracker (section 10)
Versioning	Keeps every upload; only the accepted one is used	Review (section 14); package (section 16)
Expiry watcher	Flags documents that age out	Background jobs; Action needed

8. Document security pipeline

In plain words: uploaded files could carry viruses or be fake file types, and they hold very sensitive data. So every file goes through eight automatic steps before any staff member sees it. The applicant only sees "Checking..." for a few seconds.

document security pipeline · eight automatic steps

What each step does

Step	What happens	Why
1 Signed upload link	The browser gets a one-time link valid 5 minutes, for one slot only	The file goes straight to storage; our servers never handle the raw upload
2 Quarantine store	A private bucket nobody, staff included, can open	Untrusted files are isolated
3 Real file type	Reads the file's first bytes ("magic bytes") to confirm it really is a PDF or image; checks size (10 MB) and page count	Stops disguised executables and oversized files
4 Malware scan	A ClamAV worker scans the file; virus signatures update daily	Stops infected files reaching staff computers
5 Unlock and normalise	Password-protected PDFs are unlocked with the password the applicant typed, then the password is discarded; images are rotated and compressed	Reviewers and banks can open the file
6 Quality hints	Flags blur, darkness, missing pages and expiry dates (OCR extraction comes later)	Hints help the reviewer; they never decide
7 Clean store	The file moves to an encrypted bucket under a random key; quarantine copy deleted	No guessable file names; encryption at rest
8 Ready for review	Staff see it only through a watermarked, logged preview	Every view is attributable

Rules

- Files are never public; links expire in 5 minutes or less.
- Every step writes an event: who uploaded, when, result, scanner version.
- Failures delete the file and tell the applicant in plain words ("We couldn't open this file, please upload it again").

- Purge (section 43) deletes the file and every stored version, including backups as they rotate.

Feature map

Feature	What it does	Linked to
Signed upload URLs	Time-limited, slot-specific upload	Document center (section 7)
scan-document worker	Steps 3–7	Background jobs; system design doc
Watermarked preview	Shows the reviewer name and time over the document	Document review (section 14); security (section 44)
Download approval	Downloads need a second person and expire	RBAC (section 27)

9. Customer submits application

In plain words: when everything is filled in and uploaded, the applicant reviews a summary and taps Submit. From that moment the form is locked and the back office takes over. The applicant can still fix things, but only when we ask through an Action needed task.

submit · completeness check, declaration, frozen snapshot

Step by step

1. The review screen shows every section and document slot, with an "Edit" link next to each.
2. The server checks completeness: every mandatory field valid and every mandatory slot clean. If anything is missing, it's highlighted and Submit stays disabled.
3. The applicant ticks a declaration: "The details I have given are true and complete."
4. The platform freezes **snapshot v1**: a copy of all field values and document versions with a content hash, so we can always show exactly what was submitted.

5. The state moves `DOCUMENTS_PENDING` → `VERIFICATION_REVIEW` and the application enters the review queue (section 11).
6. A confirmation SMS and email go out with the reference number and "under review".
7. The form is locked. Later changes happen only through an Action needed task (section 14), each creating a new snapshot version.

Rules

- Submission is idempotent: tapping twice creates one submission.
- Consent must still be valid at the moment of submission.
- Nothing in the confirmation message reveals personal details; it carries only the reference and a sign-in link.

Feature map

Feature	What it does	Linked to
Review screen	Summary with edit links	Sections 6–7
Completeness check	Server-side validation against the product configuration	Product configuration (section 46)
Declaration	Applicant confirms accuracy; stored with time	Audit
Snapshot	Frozen, hashed copy of what was submitted	Single source of truth (section 39)
State change + queue	Moves to <code>VERIFICATION_REVIEW</code>	Back office (section 11)
Confirmation message	Sends the reference	Communications (section 42)

10. Status tracker: what the customer sees

In plain words: after submitting, the applicant can sign in any time and see a simple progress bar. It uses plain words, never our internal codes. Anything the bank does appears only after the bank confirms it, so we never tell an applicant something that isn't true.

status tracker · what the applicant sees

Label for every state

The label is calculated from the three status fields (section 39); it's never stored on its own.

Internal state	Bank status	Applicant sees	Message sent?
ENQUIRY , PROFILE_IN_PROGRESS	—	Application started: finish your details	Reminder if idle (section 26)
CONSENT_PENDING	—	Please review and give consent to continue	Yes
DOCUMENTS_PENDING	—	Documents needed: <list>	Reminder if idle
VERIFICATION_REVIEW , APPLICATION_PREPARATION , INTERNAL_REVIEW , READY_FOR_BANK	—	Under review by Loans4India	On submit only
ACTION_NEEDED	any	Action needed: <what's missing>	Yes
ON_HOLD	any	Under review by Loans4India (reason never shown)	No
SUBMITTED_TO_BANK	received	Submitted to <bank>	Yes, names the bank
SUBMITTED_TO_BANK	processing	<bank> is reviewing your application	No
SUBMITTED_TO_BANK	query	<bank> needs more information	Yes

Internal state	Bank status	Applicant sees	Message sent?
SUBMITTED_TO_BANK	in-principle	<bank> has given in-principle approval (not a final sanction)	Yes
SUBMITTED_TO_BANK	sanctioned	<bank> has sanctioned your loan; next steps are with <bank>	Yes
CLOSED	disbursed	<bank> has disbursed your loan	Yes
CLOSED	rejected	<bank> could not approve this application	Yes
CLOSED	other reasons	Closed: <withdrawn / expired / cancelled>	Yes

Language rules

Say	Never say
"Under review by Loans4India"	"Approved", "Pre-approved", "Eligible"
"Submitted to <bank>" (after the bank reference is recorded)	"Submitted" before that
"<bank> has given in-principle approval"	Calling it a sanction
"Loans4India doesn't charge you any fee"	Any fee request
Loan application	Lead

Offer terms, the Key Fact Statement and the APR are shown exactly as the bank issued them (**VERIFY** the method with each bank).

Feature map

Feature	What it does	Linked to
Progress bar	Six steps computed from status fields	Status model (section 39)
Task chips	Action needed and bank-query tasks with an upload button	Sections 14 and 20
Notifications	Messages at the marked transitions	Communications (section 42)
Second-person gate	In-principle, sanction and manual disbursal shown only after a second check	Section 19

Part 2 — Back office and bank journey

11. Back office takes over

In plain words: the moment an applicant submits, the application becomes Loans4India's job. The system runs a few automatic checks, then places the file in a work queue where staff pick it up. Nothing suspicious is ever rejected automatically; a person always looks first.

intake · automatic checks, then the review queue

Intake checks (automatic, seconds after submit)

Check	What it compares	If it matches
Duplicate	PAN hash, mobile, bank account, device across open applications	Flag → exception queue; never auto-merge or auto-reject
Consent	Consent record is valid and covers this product's bank list	If not valid → <code>CONSENT_PENDING</code> , applicant asked

Check	What it compares	If it matches
Fraud signals	Same device or IP across many applicants, sudden bursts, mismatched names	Flag → fraud review (section 35)
Completeness (again)	Server re-runs the submit check	If it fails (rare race condition) → back to the applicant

The work queues

Staff work from queues, not from lists of applicants. Each queue is a filtered view of applications by state and flags.

Queue	What's in it	Main users
New for review	VERIFICATION_REVIEW , unassigned	Ops Lead assigns
My cases	Assigned to me, any open state	Ops Executive
Responses received	Applicant answered an Action needed task	Owner
Exceptions	Duplicate and fraud flags	Ops Lead, Compliance
Ready for Bank	Approved packages waiting for portal entry	Authorised DSA Operator
With bank: stale	Submitted, no bank update in 7 days	Ops Lead
Bank queries due	Open bank queries by due date	Owner
MIS and payout exceptions	Unmatched or conflicting rows	Ops, Finance

Priority and SLA (*proposed defaults*)

- Oldest first, with an SLA clock on each application: first review within 1 working day of submit; Ready for Bank within 3 working days if no Action needed.
- Raised priority for: bank query due within 1 day; applicant responses older than 1 day; SLA breached.

- The SLA clock pauses while the application waits on the applicant (`ACTION_NEEDED`) or is on hold.

Feature map

Feature	What it does	Linked to
Intake job	Runs the checks right after submit	Background jobs; section 9
Queues	Filtered work lists with counts	Control tower (section 38)
SLA clock	Measures waiting time per state	Metrics (section 54)
Exception queue	Holds flagged files for a person	Fraud (section 35); Compliance (section 34)

12. Application assignment

In plain words: one named person owns each application from review until it goes to the bank, and then keeps following it up. Owning means being responsible for moving it forward and answering for delays. The Ops Lead hands out work; nobody picks cases at random.

assignment · one owner per application

Step by step

1. New submissions appear in the "New for review" queue with their SLA age.
2. The Ops Lead sees each executive's open cases and scope (for example, products or banks they're trained on).
3. The Ops Lead assigns an owner and a due date. The owner sees it in "My cases" with an in-app alert.
4. If the owner is away or overloaded, the Ops Lead reassigns with a reason. Every assignment change is in the timeline.

5. **Phase 2:** automatic assignment by workload and scope, with the Ops Lead able to override.

Rules

- An application has exactly one owner at a time.
- An executive sees only their own cases and the unassigned queue (section 27).
- Unassigned cases older than 4 working hours (*proposed default*) raise an alert on the Ops Lead's dashboard.
- Leavers' cases are reassigned before their access is removed (section 44).

Feature map

Feature	What it does	Linked to
Assignment action	Sets owner and due date	RBAC: Ops Lead only (section 28)
Workload view	Open cases per person, by state	Control tower (section 38)
Reassignment log	Who moved it, when, why	Audit; timeline in the workspace (section 13)
Alerts	Unassigned age, overdue due dates	Dashboard; metrics (section 54)

13. 360° application workspace

In plain words: when staff open an application, they see one screen with everything about it: who the applicant is, what they agreed to, every document, every open problem, every bank attempt and a full history. Staff never need to look anywhere else, and the buttons in the header only allow what their role permits.

workspace wireframe · header actions, panels, documents, timeline (sample data)

Panels

Panel	Shows	Who can see it
Header	Reference, product, state label, owner, SLA time left, action buttons	All staff with case access
Overview	Key facts: amount, income, employment, bank selected, attempt status	All with case access
Applicant	All details from section 6, masked by tier; "reveal" logs a view	Owner, Ops Lead, Compliance (with reason)
Documents	Slots, versions, status, watermarked preview, accept or reject	Owner, Document Reviewer, Ops Lead
Consent	Text version, bank-list version, time, revocations	All with case access (read-only)
Issues and tasks	Action needed items, bank queries, duplicate and fraud flags	Owner, Ops Lead; flags also Compliance
Preparation	Selected bank and product, selection reason, mapped fields, checklist, approver	Owner, Ops Lead
Attempts	Each bank attempt: bank, tries, operator, bank reference, evidence, outcome	Ops, Operator
Bank updates	Bank-confirmed status history with source and evidence	Ops, Operator, Finance (read)
Timeline	Every event: who, what, when, old and new values, notes	All with case access
Finance	Disbursal reference, commission rule, payout match, adjustments	Finance only

Header actions (shown only when allowed)

Action	Who	What it does
Raise Action needed	Owner	Creates a task for the applicant (section 14)
Pass review / select bank	Owner, then Ops Lead	Moves to bank selection (section 15)
Approve package	Ops Lead (not the preparer)	Ready-for-Bank gate (section 17)
Reassign	Ops Lead	Section 12
Hold / release	Ops Lead, Compliance	Pauses the case for fraud or compliance; released by someone else
Add note	All with access	Internal note in the timeline, never shown to the applicant
Close	Owner proposes, Ops Lead confirms	Closes with a reason (section 26)

Rules

- The screen is built from the single application record (section 39); no copies in spreadsheets or email.
- Every reveal of a masked field and every document view is logged.
- Buttons call server actions that check the role and state again; hiding a button is never the only control.

Feature map

Feature	What it does	Linked to
Workspace screen	One view per application	Every Part 2 section
Masking and reveal	Hides sensitive fields until needed	Data tiers (section 44)
Timeline	Human-readable audit trail	Audit log (section 39)
Transition buttons	Only valid next steps, server-checked	Status model (section 39)

14. Document and data review

In plain words: the owner checks two things. First, is each document usable: readable, the right type, not expired, complete? Second, does everything agree: names, dates, employer, salary across the form and the documents? Anything wrong goes back to the applicant as a clear task. This is a quality check so the bank gets a clean file. It is **not** KYC and never a credit judgement.

review · document quality, then data consistency

What the reviewer checks per document

Document	Checks
PAN card	Readable; name and DOB match the form; PAN number matches the form
Address proof	Readable; name matches; address matches the form; within validity (utility bill under 3 months)
Salary slips	Last 3 consecutive months; employer name matches; net pay roughly matches stated income
Bank statements	Full 6-month period, no gaps; account holder name matches; salary credits match the slips; no signs of editing

Document	Checks
Business documents	GST, ITR or registration names and numbers match the form
Masked Aadhaar (if used)	First 8 digits hidden; name and address match

Data checks across the file

- **PAN/name match** through a licensed provider: result stored as match, partial or no match, with the provider's reference.
- Same name and DOB everywhere; employer the same on form, slips and statements; stated income within a tolerance of documented income (10%, *proposed default*).
- Tampering signals from section 35 (edited PDFs, mismatched fonts or totals) go to fraud review, not to the applicant.

Reason codes for rejecting a document

BLURRY · UNREADABLE · WRONG_DOCUMENT · EXPIRED · MISSING_PAGES · NAME_MISMATCH · PERIOD_INCOMPLETE · PASSWORD_PROTECTED · OTHER (free text required)

The Action needed task

1. The reviewer picks the reason and writes a short instruction in plain words ("Please upload your May salary slip; the one we have is cut off at the bottom").
2. The application moves to ACTION_NEEDED ; the applicant gets an SMS and email with a sign-in link (no personal details in the message).
3. Only the slots or fields named in the task unlock for editing.
4. The applicant responds; a new snapshot version is frozen and the case returns to the owner's "Responses received" queue.
5. Reminders at 2 and 5 days; the application closes as expired at 15 days without a response (*proposed defaults*, section 26).

Rules

- The reviewer can't mark anything as "KYC complete" or "approved".
- Rejected versions stay in history and can't go to a bank.
- Staff can correct obvious typos in non-identity fields with a reason; identity fields (name, DOB, PAN, mobile) change only from the applicant.

Feature map

Feature	What it does	Linked to
Document decision	Accept, or reject with a reason code	Document center (section 7)
PAN/name match	Licensed check, stored as a result plus reference	Integrations; fraud (section 35)
Consistency checks	Compares fields with documents	Workspace (section 13)
Action needed task	Asks the applicant to fix specific items	Status tracker (section 10); communications (section 42)
Expiry timers	Remind, then close	Section 26

15. Bank selection and sourcing-criteria check

In plain words: once the file is clean, Loans4India decides which bank to send it to. The applicant doesn't choose. The system first removes banks that can't take this file: not on the list the applicant agreed to, no active contract for this product, or outside the bank's own sourcing rules. The Ops Lead then picks one of the remaining banks and writes down why. This replaces the old "Lender eligibility check" and "Lender assignment"; it is not a credit decision.

bank selection · filters, candidates, one pick

The three filters

Filter	Removes a bank when	Data it uses
On consented list	The bank isn't on the bank-list version the applicant consented to	Consent record (section 5)
Active agreement	No signed agreement or DSA code today, or the product isn't covered	Bank configuration (section 47)

Filter	Removes a bank when	Data it uses
Sourcing rules	The applicant is outside the bank's published sourcing criteria: city served, age range, minimum income, employment type, employer category	Bank configuration: sourcing rules

Step by step

1. When review passes, the owner clicks "Pass review"; the case appears in the Ops Lead's "Select bank" view.
2. The platform runs the three filters and lists the candidate banks with the reason any others were removed.
3. The Ops Lead picks one and records the basis: product coverage, sourcing fit, bank turnaround or capacity.
4. A **submission attempt** is created for that bank (attempt state `PREPARING`); the application moves to `APPLICATION_PREPARATION`.
5. Only one attempt can be active per application.

Edge cases

Situation	What happens
No bank passes the filters	Close as "no eligible bank", with a plain message to the applicant
A suitable bank isn't on the applicant's consented list (added later)	One-tap re-consent request (<code>CONSENT_PENDING</code>); if declined, pick another or close
Agreement expires mid-way	The filter fails at the gate (section 17); pick again
Re-routing after a rejection	Same filters, minus banks already tried (section 21)

Rules

- Never an approval-probability score or ranking of applicants. Selection must be explainable from the recorded basis (**VERIFY** unbiased-display rules if several banks are ever shown to applicants).
- The applicant sees the bank's name only after submission (section 18).

Feature map

Feature	What it does	Linked to
Bank filters	Remove banks that can't take the file	Consent; bank configuration; empanelment (section 47)
Selection record	Bank, product, reason, who, when	Attempt record (section 39)
Re-consent request	Approve an off-list bank	Consent (section 5)
Attempt creation	Starts the one-bank submission	Sections 16–18

16. Submission package

In plain words: every bank wants the same information arranged its own way: different field names, date formats, document names and order. A **bank template** describes one bank's format. The platform combines our single application record with that template to produce a ready-to-enter package, so the operator types exactly what the bank expects and nothing is missing.

submission package · record + template → manifest

What's in a package

Item	Description
Mapped field values	Every field the bank's portal asks for, already converted to its format
Documents	The accepted version of each required document, renamed to the bank's convention (for example <code>PAN_{ref}.pdf</code>)
Application summary	A generated PDF summarising the file, for banks that want one
Cover checklist	Each requirement with a tick and its source
Manifest	Template version, field values, document version IDs and a content hash, so we can prove later exactly what was sent

Field mapping example

Our field	Bank's field	Conversion
<code>employment_type = SALARIED</code>	"Occupation"	"Salaried" (from a value map)
<code>dob = 1994-03-07</code>	"Date of Birth"	<code>07/03/1994</code>
<code>monthly_income = 85000</code>	"Annual Income"	$\times 12 = 1,020,000$
<code>residence_type = RENTED</code>	"Residence"	"Rented"
<code>loan_purpose = MEDICAL</code>	"End use"	"Medical expenses"

Step by step

1. On creating the attempt (section 15), the platform loads the bank's **active** template version.
2. The mapping engine fills every bank field from our record, applying the conversions.
3. Any required bank field with no value, or any document slot without an accepted version, makes the checklist fail.

4. The owner fixes what we can (mapping, a typo with reason) or raises an Action needed task for the applicant.
5. When the checklist is 100%, the owner sends the package to the Ready-for-Bank gate (section 17).

Rules

- Templates are versioned and never edited in place; a change creates a new version that needs approval (section 46).
- The package always uses accepted, unexpired document versions only.
- Bank-specific extra questions are added to the application form through the template, so the applicant is asked up front (section 6).

Feature map

Feature	What it does	Linked to
Bank template	Field map, value maps, document list, naming rules	Product catalogue and bank configuration (section 46)
Mapping engine	Converts our record to the bank format	Application record (section 39)
Checklist	Proves completeness	Gate (section 17)
Manifest and hash	Frozen proof of what was prepared	Portal entry (section 18); audit

17. Ready-for-Bank gate

In plain words: this is the last checkpoint before anything goes to a bank. The system re-checks eight things at that moment, and a **different person** from the one who prepared the package must approve it. Only then is the package frozen and handed to the operator who will enter it on the bank's portal. MVP1 ends here.

Ready-for-Bank gate · eight automatic checks plus a second person

The checks

#	Check	Why
1	Consent is valid right now	The applicant may have revoked it since submitting
2	The bank is on the applicant's consented bank-list version	Data may only go to a named bank
3	The bank's agreement and DSA code are active today	Contracts can expire
4	Every required bank field has a value	The portal will reject gaps
5	Every required document has an accepted, unexpired version	Old statements get returned
6	No open Action needed task, hold, duplicate or fraud flag	Only clean files go out
7	The template version is recorded in the manifest	Proof of what format was used
8	The approver is not the preparer	Four-eyes control

Step by step

1. The owner sends the package to approval. The Ops Lead (or a named second reviewer) sees it in their queue.
2. The approver opens the package preview, checks the cover checklist, and clicks Approve, or Send back with a reason.
3. On approval the server runs all eight checks again. Any failure blocks approval and shows which check failed.
4. On success: application `READY_FOR_BANK` , attempt `READY_TO_SUBMIT` , package frozen with its hash, and the case appears in the operator queue.
5. If anything changes after approval (a document expires, consent is revoked), the case returns to preparation automatically.

Rules

- The gate is enforced on the server; the Approve button being visible isn't enough.

- For a very small team, a named outside approver (for example, the founder) can hold an approval-only account; self-approval is never allowed (section 27).
- The applicant still sees "Under review by Loans4India".

Feature map

Feature	What it does	Linked to
Gate service	Runs the eight checks at approval time	Consent; bank configuration; documents; review
Approval queue	Packages waiting for a second person	Ops Lead (section 28)
Package freeze	Locks values and document versions with a hash	Portal entry (section 18)
Auto-unfreeze	Sends the case back if a check stops being true	Background jobs

18. Bank portal entry

In plain words: This is the moment the application actually leaves Loans4India and reaches the bank. Banks give a DSA (Direct Selling Agent) a login to their own website, the *DSA portal*. A named person at Loans4India, the **Authorised DSA Operator**, logs in with their own personal credentials and types the application into the bank's portal by hand, uploading the documents we prepared. Nothing is sent automatically, and no robot (RPA) fills the portal. When the bank's portal gives back a reference number, the operator records it in our system, and from that point the application is "with the bank".

Figure 18 — Bank portal entry: from frozen package to bank reference, with retr...

Step by step

1. **Pick up from the queue.** The operator's queue lists attempts in `READY_TO_SUBMIT`, oldest first, with the selected bank, product and SLA timer. Only operators who hold an active portal credential for that bank see that bank's items.

2. **Open the frozen package.** The package approved at the Ready-for-Bank gate (section 17) opens read-only. Its hash is shown; if anything changed since approval, the screen blocks entry and sends it back to the gate.
3. **Download documents.** Each file is fetched through a short-lived signed link (default 10 minutes, proposed). Every download is logged with who, when, file and purpose "bank submission". Files are watermarked "For <Bank> submission only" where the bank accepts it.
4. **Enter on the bank portal.** The operator works in a separate browser tab, on the bank's site, with their own login. Our platform never stores the bank portal password (it lives in the company password vault, outside the product).
5. **Record the result.** The operator pastes the bank's application/lead reference, uploads a screenshot of the portal confirmation page, picks the bank branch or hub if shown, and ticks "I submitted this personally".
6. **System commits.** In one database transaction: attempt → SUBMITTED , application → SUBMITTED_TO_BANK , bank-confirmed status → RECEIVED (source: portal), plus an audit event. If any part fails, nothing changes.
7. **Notify the applicant.** SMS/email/WhatsApp (DLT-registered template): "Your application has been submitted to <Bank>. Reference: <our ID>." This is the disclosure that replaces a pre-submission notice (VERIFY with counsel).

Rules

- Only the Authorised DSA Operator role can mark a submission. The operator cannot be the same person who prepared the package (separation of duties).
- A bank reference is mandatory and must be unique per bank. Duplicate references are rejected.
- Screenshot evidence is mandatory. It is stored in the clean store at tier T2.
- No bulk download ("zip all") outside this screen. No copy to personal devices; downloads go to a managed machine only (policy, VERIFY with IT controls).
- Submission must happen within 2 working days of Ready for Bank (proposed SLA); after that the item turns amber, then red, and the Ops Manager is alerted.
- A consent check runs again at the moment of entry: if the applicant has withdrawn consent or the bank is no longer on their consented list, the screen blocks.

Feature map

Feature	What it does	Linked to
Operator queue	Lists READY_TO_SUBMIT attempts per bank	Section 17 gate, section 30 operator role
Frozen package viewer	Read-only package with integrity hash	Section 16 package, audit log
Signed download link	Expiring, logged file access	Section 8 document store, section 44 security
Submission record form	Bank ref, screenshot, branch, attestation	Attempt table, bank-status events
Transactional status commit	Updates three status fields atomically	Section 39 status model
Consent re-check	Blocks entry if consent changed	Section 5 consent ledger
Applicant notification	Tells the applicant which bank received it	Section 42 communications
Retry and escalation	Handles portal failures safely	Ops Manager queue, Bank/Partner Manager

Edge cases

- **Portal down or times out mid-entry.** Operator marks "Entry failed" with a reason. Before retrying, they must search the bank portal for a partial or duplicate lead ("partial-entry check") and record what they found, so the same applicant is never submitted twice. After 3 failed tries the item escalates.
- **Bank rejects at the portal (e.g. pin code not serviced).** Recorded as attempt `RETURNED_OR_FAILED` with the portal message; goes back to bank selection (section 15), which counts towards the re-route limit.
- **Operator's portal credential expires or is revoked.** The queue hides that bank's items for them; the Bank/Partner Manager is alerted.
- **Wrong bank reference typed.** Correction is a new superseding event with a reason and a second-person check, never an edit in place.

19. Bank processing and tracking

In plain words: After submission, the bank does its own work: verification calls, credit checks, property or field visits, and its own approvals. We cannot see inside the bank. What we can see is *evidence the bank sends us*: the daily or weekly **MIS file** (a spreadsheet listing every lead we sourced and its stage), API updates for the few banks that offer them, emails from the bank's sales manager, or what the operator sees on the portal. This section is the machinery that turns that evidence into our **bank-confirmed status**, safely and without guesswork.

Figure 19 — Bank processing and tracking: evidence in, match, precedence, check...

Step by step

1. **Every submitted attempt gets an owner and a next follow-up date.** The case owner (Loan Ops Executive) sees it in "With bank" on their queue.
2. **Evidence arrives.** MIS files are uploaded by the Bank/Partner Manager (or fetched from a shared mailbox in Phase 2). Each upload is stored as a raw file, hashed and never edited.
3. **Import and map.** Each bank has a *column map*: which column holds the bank reference, stage, amount, date. Maps are versioned; when a bank changes its MIS layout, a new map version is created and approved, and old files still read with their old map.
4. **Match.** Each row is matched on bank reference → attempt → application. If the reference is missing or unknown, the row goes to the **exception queue**; Ops can match by hand with a reason, and the match is logged.
5. **Translate the bank's words.** Each bank uses its own stage names ("Login done", "Credit approved", "PDD pending"). A per-bank *stage dictionary* maps them to our nine bank statuses: RECEIVED, PROCESSING, QUERY, IN_PRINCIPLE, SANCTIONED, AGREEMENT_COMPLETED, DISBURSED, REJECTED, CANCELLED. Unknown stage words go to the exception queue; they are never guessed.
6. **Apply source precedence.** If two sources disagree, the stronger one wins: bank API > bank MIS > bank email > portal screenshot > master-DSA report. A weaker source can never overwrite a stronger one for the same date.
7. **Checker for key milestones.** IN_PRINCIPLE, SANCTIONED and any *manually entered* DISBURSED need a second person to confirm against the evidence before they show to the applicant or count for commission.

8. **Save the event.** A new bank-status event is appended with source, evidence file, row number and who confirmed it. The applicant's tracker updates and a notification goes out where the template allows.

Rules

- Bank status is set **only** from bank evidence. Nobody can type "Sanctioned" because a bank officer said so on the phone, unless they attach the email or screenshot.
- Status events are append-only. A wrong status is fixed with a *superseding event* (reason + checker), never by editing or deleting.
- Status cannot move backwards without a reason (e.g. SANCTIONED → CANCELLED is allowed; SANCTIONED → RECEIVED is flagged).
- Stale rule: no bank evidence for 7 days (proposed) creates a follow-up task. 30 days (proposed) escalates to the Bank/Partner Manager.
- MIS files contain other people's data (the bank's full lead list). Rows that do not match our attempts are discarded after the exception window and never stored as customer records.

Feature map

Feature	What it does	Linked to
MIS upload	Stores the raw bank file with hash	Bank/Partner Manager role, storage
Column map (versioned)	Reads each bank's file layout	Bank master (section 47)
Stage dictionary	Bank words → our nine statuses	Section 39 status model
Matcher	Bank ref → attempt → application	Attempt table, section 18 bank ref
Exception queue	Holds unmatched or unknown rows	Ops Manager dashboard (section 38)
Source precedence	Stops weaker sources overwriting stronger	Status engine
Milestone checker	Second person on key statuses	Separation of duties, audit log
Stale timer	Follow-up tasks after silence	Job scheduler, notifications

Feature	What it does	Linked to
Applicant tracker update	Shows the new status in plain words	Section 10 status tracker

Edge cases

- **Same row in two MIS files.** The matcher sees the same bank ref + stage + date and ignores the duplicate (idempotent import).
- **Bank reference changes inside the bank** (e.g. lead ID becomes loan ID). Ops links the new reference to the attempt as an alias, with evidence.
- **MIS shows a stage we never got** (e.g. jumps from RECEIVED to SANCTIONED). Allowed; intermediate states are not invented.
- **MIS shows an applicant we never submitted.** Goes to exception queue and is discarded; it may be a lead from another DSA.

20. Bank query handling

In plain words: While processing, the bank often asks for something more: a missing bank statement, a clearer address proof, an explanation of a cheque bounce, a co-applicant's PAN. This is called a **query** (bank status **QUERY**). Our job is to capture exactly what the bank asked, get the answer quickly, and send it back through the bank's own channel with proof that we did.

Figure 20 — Bank query: log, decide who answers, reply through the bank's chann...

Step by step

1. **Query arrives** via MIS row, email from the bank, or a flag seen on the portal. It is linked to the attempt and the bank status becomes **QUERY**.
2. **Log the exact ask.** The case owner copies the bank's wording word for word, attaches the evidence, sets the due date (from the bank, or 3 working days by default, proposed) and owner.
3. **Decide who answers.** If we already hold the answer (e.g. a document we collected but the bank missed), Ops answers from the case file. If only the applicant can answer, an **applicant task** is created.

4. **Applicant task.** The applicant gets a notification with a secure link that shows a plain-words version of the request ("Please upload your last 3 months' salary slips"). They do not see the bank's internal remarks. Uploaded files go through the full security pipeline (section 8) and are reviewed by the Document Reviewer.
5. **Reply via the bank channel.** The operator uploads the answer on the portal or replies to the bank's email, and saves proof (screenshot or sent email) on the query.
6. **Close the query** on our side as "Answered". The bank status stays **QUERY** until bank evidence moves it on (section 19).

Rules

- The application state becomes **ACTION_NEEDED** only while an applicant task is open; otherwise it stays **SUBMITTED_TO_BANK**.
- New documents collected for a query must be covered by the original consent purpose. If the bank asks for something outside it (e.g. a family member's data), fresh consent is needed from that person.
- Unanswered queries escalate to the Ops Manager 1 day before the due date; overdue queries show red on the control tower.
- Applicant reminders: day 1, day 3, day 6 (proposed), then the case owner calls.

Feature map

Feature	What it does	Linked to
Query record	Bank's exact ask, due date, owner, evidence	Attempt, bank status QUERY
Applicant task	Plain-words request with secure upload	Sections 7–8 document center and pipeline
Reply evidence	Proof the answer reached the bank	Audit log
Query SLA clock	Reminders and escalation	Control tower (section 38)

Edge cases

- **Applicant does not respond.** After the reminders and a call, the case owner records "Applicant unresponsive". If the bank closes the lead, bank status becomes CANCELLED from its evidence.
- **Query asks for something the applicant cannot provide.** Recorded as a reason; the Bank/Partner Manager may ask the bank to proceed or the case may be re-routed (section 21).
- **Multiple queries open at once.** Each is its own record with its own clock; the applicant sees one combined checklist.

21. Bank decision (approval, rejection, re-route)

In plain words: Sooner or later the bank decides. It may give an **in-principle** approval ("we will lend, subject to conditions such as a property valuation"), a full **sanction** (a firm offer with amount, rate and tenure), a **rejection**, or the lead may be **cancelled** (by the bank, or because the applicant withdrew). Our job is to record the decision exactly as the bank gave it, and — if the bank said no for a reason that another bank might see differently — to try one more bank from the applicant's consented list.

Figure 21 — Bank decision: approval path, rejection, and the re-route rule

Step by step

1. **Decision arrives as evidence** (section 19). In-principle and sanction need a checker before they become visible.
2. **Approved path.** IN_PRINCIPLE → SANCTIONED. We record the sanctioned amount, rate and tenure *only if the bank shares them*, as "bank-reported" fields. The attempt becomes `OUTCOME_RECEIVED`.
3. **Rejected path.** The rejection reason is recorded in the bank's own words plus a reason code from our list (e.g. policy: pin code, policy: employer category, credit, income, documents, fraud suspicion, other).
4. **Re-route decision.** The Ops Manager decides whether the reason is *bank-specific* (another bank's policy could differ: pin code, employer category, product limits) or *applicant-wide* (would fail anywhere: fraud flags, forged documents, applicant withdrew).

5. **If re-routed:** a new attempt is created for a different bank from the consented list, the old attempt becomes `SUPERSEDED`, and the case returns to bank selection (section 15) and the full gate again (section 17). The applicant is told "<Bank A> could not approve; we are now submitting to <Bank B>".
6. **If not re-routed:** the application closes with reason `REJECTED` (section 26) and the applicant is told in plain, respectful words. We do not share internal fraud flags.
7. **Cancelled:** bank status `CANCELLED` from evidence; the application closes with reason `CANCELLED_BY_BANK` or `WITHDRAWN`.

Rules

- Maximum 2 re-routes per application (proposed), so at most 3 banks ever see one application.
- Re-routing only to banks on the applicant's consented list. A bank outside the list needs one-tap re-consent first (section 5).
- A rejection for suspected fraud or forged documents is **never** re-routed; it goes to the Fraud & Risk queue (section 35).
- Approval details (amount, rate) are shown to the applicant labelled "as reported by <Bank>". We never present them as our offer.

Feature map

Feature	What it does	Linked to
Decision capture	Records in-principle, sanction, rejection, cancel	Section 19 evidence, status model
Reason codes	Standard list + bank's own words	Metrics (section 54), bank scorecards
Re-route action	New attempt, old <code>SUPERSEDED</code> , limit 2	Sections 15–17, consent ledger
Applicant messaging	Plain-words outcome per path	Section 42 communications
Fraud hand-off	Blocks re-route, opens fraud case	Section 35 Fraud & Risk

Edge cases

- **Sanction then cancelled** (e.g. applicant refuses the rate). Recorded as CANCELLED; no re-route unless the applicant asks and Ops agrees.
- **Bank reverses a rejection on appeal.** A superseding event with evidence; the attempt continues.
- **Two banks approve** (only possible if a re-route happened after a slow, not final, response). Not allowed by design: a re-route requires the earlier attempt to be closed first.

22. Loan agreement and Key Fact Statement

In plain words: After sanction, the bank and the applicant sign the loan agreement. The bank must give the applicant a **Key Fact Statement (KFS)** — a standard summary showing the loan amount, interest, all fees and the **APR** (Annual Percentage Rate, the all-in yearly cost). This is entirely between the bank and the applicant. Loans4India is not a party to the loan. We only record that it happened, when the bank tells us.

Figure 22 — Agreement: bank-led steps and what we record

Step by step

1. Bank issues the sanction letter and KFS to the applicant directly.
2. The applicant reviews and signs with the bank (branch, e-sign or bank app).
3. When bank evidence shows the agreement is done, we record `AGREEMENT_COMPLETED` (source + evidence).
4. The applicant's tracker shows "Loan agreement completed with <Bank>".
5. Optional: if the bank shares a copy of the KFS with us and the applicant consented to it, we store it at tier T2 for support questions.

Rules

- We never generate, edit or explain loan terms as if they were ours. Support answers use the bank's KFS wording, and direct rate or fee disputes to the bank's grievance channel.

- We never collect money from the applicant, at any stage (no processing fee, no "file charge"). Any staff request for money is a disciplinary and fraud matter.
- If the bank does not report this stage, it is simply skipped; disbursement can follow directly.
- VERIFY with counsel: whether, as DSA, we must also hand over or confirm receipt of the KFS for any partner bank.

Feature map

Feature	What it does	Linked to
Agreement status	Records AGREEMENT_COMPLETED from evidence	Section 19 evidence engine
KFS copy (optional)	Stores bank-issued KFS for support	Data tiers (section 43), consent
No-fee guard	Support scripts + fraud alert keywords	Sections 36 support, 35 fraud

Edge cases

- **Applicant asks us to negotiate the rate.** Support explains we can pass the request to the bank's sales manager but cannot promise anything; logged as a note.
- **Applicant refuses to sign.** Bank evidence will show CANCELLED; handled as in section 21.

23. Disbursement

In plain words: Disbursement is when the bank actually pays the loan money — into the applicant's account, to the seller of a property, or to a car dealer. It is the finish line for the applicant, and it is also the event that earns Loans4India its commission. Because money depends on it, we only mark an application as disbursed when the bank's own evidence says so.

Figure 23 — Disbursement: evidence, checker for manual entries, close and hand ...

Step by step

1. **Bank disburses** (full or first tranche).
2. **Evidence arrives** as an MIS row or API update with disbursed amount, date and the bank's loan account number.
3. **Automatic or manual.** A matched MIS/API row records directly. A manual entry (e.g. from a bank email) needs a checker who compares it with the attached evidence.
4. **Record.** Bank status **DISBURSED** ; attempt **CLOSED** ; disbursed amount, date and loan account number (masked except last 4 digits) saved.
5. **Close the application** with reason DISBURSED (section 26).
6. **Notify the applicant:** "Your loan from <Bank> has been disbursed." with a link to the bank's grievance contact.
7. **Create the expected commission** entry (section 24).

Rules

- Tranches: home and business loans may disburse in parts. The first tranche closes the application; later tranches are logged as additional disbursal events on the closed attempt and each creates its own expected commission (if the schedule pays on tranches).
- Disbursed amount must be $\leq$ the sanctioned amount if we know it; otherwise flagged for review.
- A disbursal reversal (cancelled after payment) is a superseding event and creates a commission clawback.
- Loan account numbers are T2 data and are masked on all screens except Finance.

Feature map

Feature	What it does	Linked to
Disbursal record	Amount, date, loan account, source	Section 19 evidence, attempt
Manual-entry checker	Second person for non-MIS evidence	Separation of duties
Tranche log	Multiple disbursal events per loan	Commission engine (24)
Close-out	Closes application as DISBURSED	Section 26 closing

Feature	What it does	Linked to
Applicant notice	Final message + bank grievance link	Section 42 communications

Edge cases

- **Disbursal shows in MIS before sanction.** Accepted (banks report late); missing states are not invented.
- **Amount differs between two sources.** The higher-precedence source wins; the conflict is logged for the Bank/Partner Manager.
- **Applicant says money not received but bank says disbursed.** Support points them to the bank; we cannot change the status without bank evidence.

24. Commission engine

In plain words: Banks pay a DSA a **commission** (also called payout) for every loan that disburses, usually a percentage of the disbursed amount, sometimes with slabs (higher rate above a volume), caps, or bonuses. The rates are written in our DSA agreement with each bank. The commission engine reads that agreement as a **schedule** and, for every disbursal, calculates what the bank *should* pay us. It does not move money; it creates an expected entry that Finance later matches against what the bank *actually* pays (section 25).

Figure 24 — Commission engine: disbursal to expected receivable

Step by step

1. **Trigger:** a confirmed disbursal or tranche event.
2. **Find the schedule** in force on the disbursal date for that bank + product (+ channel, e.g. via master DSA). Schedules are versioned with effective-from and effective-to dates.
3. **Calculate:** base = disbursed amount × rate, then slabs, caps, minimums, and any deductions written in the schedule. GST treatment and TDS (section 194H, VERIFY rate) are calculated as separate lines.
4. **Post an expected entry** to the commission ledger: bank, application, attempt, schedule version, amount, tax lines, status **EXPECTED**.

5. **Wait for payout** — reconciliation (section 25) moves it to MATCHED, SHORT, EXCESS or DISPUTED.

Rules

- Schedules are entered by the Bank/Partner Manager and approved by Finance (maker-checker) before they apply. A schedule cannot be edited after it has been used; changes create a new version.
- The ledger is **append-only**: corrections are reversal + new entry, with reason and approver.
- If no schedule matches, the entry is created with amount 0 and flagged "Schedule missing" — the disbursal is never lost.
- Volume slabs that depend on a month's total are recalculated at month end as an adjustment entry, not by rewriting earlier entries.
- MVP1 note: the engine is Phase 2. MVP1 stops at Ready for Bank, but the schedule and ledger tables are designed now so nothing needs reworking.

Feature map

Feature	What it does	Linked to
Schedule library	Versioned commission rules per bank/product	Bank master (47), Finance approval
Calculator	Amount, slabs, caps, tax lines	Disbursal events (23)
Commission ledger	Append-only expected, actual, adjustments	Reconciliation (25), audit
Missing-schedule alert	Flags zero-amount entries	Finance queue
Clawback	Negative entry on disbursal reversal	Section 23 reversal

Edge cases

- **Schedule changed mid-month.** Each disbursal uses the version in force on its own date.
- **Sourced via a master DSA** (we sit under another DSA). The schedule records the master DSA's share; expected entry is our net.

- **Bank pays a one-off incentive** not in any schedule. Finance records it as a manual income entry with evidence and approval.

25. Finance reconciliation

In plain words: Banks usually pay DSA commission monthly, in one lump sum, with a **payout file** listing which loans the money is for. Payout files are often incomplete or late, rates are applied wrongly, and some loans are missed. **Reconciliation** means matching every line the bank paid to the expected entries our engine created (section 24), and chasing every difference. It is how the business makes sure it is paid in full.

Figure 25 — Finance reconciliation: payout file to settled ledger

Step by step

1. **Upload** the payout file and the bank remittance advice (amount received in our bank account). Stored raw, hashed, never edited.
2. **Match** each payout line to an expected entry using loan account, bank reference or applicant reference + disbursal date.
3. **Classify** each line: **MATCHED** (amount within tolerance, e.g. $\pm ₹1$, proposed), **SHORT** (paid less), **EXCESS** (paid more), **UNMATCHED** (paid for a loan we cannot find), **DUPLICATE** (paid twice). Expected entries with no payout after the bank's cycle become **OVERDUE**.
4. **Resolve exceptions.** Finance raises a dispute with the bank (with our evidence) or posts an adjustment. Every adjustment needs a second approver.
5. **Tax lines.** TDS deducted by the bank under section 194H is recorded against the payout and reconciled with Form 26AS / the bank's TDS certificate. GST invoices to the bank are issued per our tax setup (VERIFY with the CA).
6. **Month-end close.** Open items are aged (0-30, 31-60, 60+ days) and reported; nothing is written off without Finance lead approval.

Rules

- Finance can see loan account numbers and amounts but not applicant documents.
- The person who uploads a payout file cannot approve adjustments on it.

- The ledger is append-only; a reconciled period is locked, and late changes go into the next open period.
- Remittance total must equal the sum of payout lines $\pm$ tax; a mismatch blocks matching until explained.

Feature map

Feature	What it does	Linked to
Payout import	Raw file + remittance, per-bank column map	Bank master (47)
Matcher + classifier	Matched / short / excess / unmatched / duplicate / overdue	Commission ledger (24)
Dispute tracker	Claims raised with banks, evidence, status	Bank/Partner Manager (32)
Adjustment approval	Maker-checker on every adjustment	Finance role (33), audit
Tax register	TDS 194H, GST lines and certificates	Legal & tax (section 50)
Ageing report	Open items by age bucket	Metrics (54)

Edge cases

- **Bank pays for a loan that a different DSA sourced.** UNMATCHED → returned or held; never booked as income without evidence.
- **Bank claws back a past payout** (loan foreclosed early or reversed). Recorded as a negative payout line matched to the original entry.
- **Payout received but no file.** Booked as "unallocated receipt" until the file arrives.

26. Application closed, withdrawal and data purge

In plain words: Every application must end in one clear place: **CLOSED** with a **close reason**. Closing is not the end of our responsibility — we still hold the applicant's personal data, and the law (DPDP Act) and our bank agreements say we may keep it only as long as needed. So closing starts a **retention clock**; when it runs out, a scheduled **purge job** deletes or anonymises the data and writes a log proving it did.

Figure 26 — Closing: four ways to close, one closed state, then retention and p...

Close reasons

Reason	When	Who sets it
DISBURSED	Bank evidence of disbursal	System (from section 23)
REJECTED_BY_BANK	Bank rejection, no re-route	Ops Manager
REJECTED_INTERNAL	Fails our review (fraud, forged docs, out of scope)	Ops Manager, with reason code
CANCELLED_BY_BANK	Bank evidence of cancellation	System
WITHDRAWN	Applicant withdraws	Applicant (self-service) or Support on request
CONSENT_REVOKED	Applicant withdraws consent	Applicant
DORMANT	No applicant activity past the timers	System job
DUPLICATE	Same applicant + product already open	Ops

Withdrawal and revocation

- **Before submission:** the application closes at once. No data has gone to a bank. Purge is scheduled.
- **After submission:** we close our side, tell the applicant we will ask the bank to stop processing, and the operator sends that request via the bank's channel (evidence saved). The bank's own data retention is the bank's responsibility, which we explain in plain words.
- Consent revocation also stops all marketing and contact, except messages required by law (e.g. confirming the revocation).

Dormancy timers (proposed)

- Incomplete application (before submit): reminder after **7 days** of inactivity, auto-close as DORMANT after **30 days**.
- **ACTION_NEEDED** with no response: reminders, then close after **15 days** if not yet submitted to a bank. After submission, dormancy does not apply; the bank status decides.

Retention and purge

1. On close, each data class gets a purge date from the retention table (documents, profile, consent records, audit events, commission records have different periods; VERIFY each with counsel and the bank agreements).
2. The nightly purge job deletes documents from storage, removes or anonymises personal fields, and keeps only what must be kept (consent records, audit events, financial records for tax law).
3. A **purge log** entry records what was purged, when and by which job version — with no personal data in it.
4. A legal hold (dispute, fraud case, regulator request) pauses the purge for that application until released by Compliance.

Feature map

Feature	What it does	Linked to
Close action	Sets CLOSED + reason, audit event	Status model (39)
Self-service withdraw	Applicant closes their own application	Section 10 tracker, section 41 data rights
Dormancy job	Reminders and auto-close	Job scheduler, communications (42)
Retention table	Purge period per data class	Data governance (43)
Purge job + log	Deletes / anonymises, proves it	Compliance (34), audit
Legal hold	Pauses purge	Compliance, Fraud (35)

Edge cases

- **Applicant returns after DORMANT close.** They start a new application; the old one is not reopened (consent may have changed).
- **Bank asks for a document after we purged it.** We tell the bank it is no longer held; the bank asks the applicant directly.
- **Commission still pending on a closed application.** Financial records are kept for the tax retention period; only personal fields are removed.

Part 3 — Admin roles

27. Roles overview (who does what)

In plain words: A **role** is a job inside Loans4India with a fixed set of screens and actions. Every staff login has exactly one role (plus, rarely, a time-limited extra permission). Roles exist for two reasons: so each person sees only the customer data their job needs (privacy), and so no single person can both do and approve sensitive work (**separation of duties**, or SoD). This part describes each role with a *Sees / Does / Cannot* card that the development team can turn directly into permission rules.

Figure 27 — Role map: ten roles, four groups, MVP1 roles highlighted

How roles work (for developers)

1. **RBAC** (role-based access control): each screen and API action checks a permission like `application.assign` or `gate.approve`. Roles are bundles of permissions. The full matrix lives in the RBAC & PAM Spec.
2. **Scope**: a permission can be limited to "own cases", "team cases" or "all". Example: a Loan Ops Executive can edit only applications assigned to them.
3. **PAM** (privileged access management): high-risk actions — role changes, data exports, viewing full Aadhaar-masked documents in bulk, purge overrides — need a time-limited elevation with a reason and an approver, and are recorded.
4. **SoD rules** are enforced by the system, not by trust: preparer $\neq$ gate approver; uploader of a payout file $\neq$ approver of its adjustments; requester of elevated access $\neq$ its approver.
5. **MFA** for every staff login; sessions time out after 15 minutes idle (proposed).

MVP1 vs later

MVP1 runs on synthetic data and stops at Ready for Bank, so it needs four logins: **Super Admin, Ops Manager, Loan Ops Executive** (who also performs document review in MVP1) and **Compliance & Audit** (proposed). The DSA Operator, Bank/Partner Manager, Finance, Fraud & Risk and Customer Support roles are switched on in Phase 2, when the first bank agreement is signed. The data model and permission names for all ten roles are built in MVP1 so nothing has to be redesigned.

Role summary

Role	One-line job	Phase
Ops Manager	Runs the queues, approves at the gate	MVP1
Loan Ops Executive	Works each application end to end	MVP1
Document Reviewer	Checks each document	MVP1 (done by Ops Exec)
Authorised DSA Operator	Enters on the bank portal	Phase 2
Bank/Partner Manager	Owens bank relationships, MIS, bank master	Phase 2
Finance	Commission, payouts, tax	Phase 2
Compliance & Audit	Consent, audit log, data rights, purge	MVP1
Fraud & Risk	Fraud flags and investigations	Phase 2
Customer Support	Answers applicants	Phase 2
Super Admin	Users, roles, configuration	MVP1

28. Ops Manager

In plain words: The Ops Manager is the shift supervisor of the application factory. They make sure every application has an owner, nothing is stuck, and only good packages go to a bank. They are the main **approver** at the Ready-for-Bank gate (section 17) and decide hard cases such as re-routing or internal rejection.

Figure 28 — Ops Manager: sees, does, cannot

A day in the life

- 1. **Morning:** opens the control tower (section 38): new applications overnight, SLA breaches, cases with no owner, exception queues.
- 2. **Assigns** unowned cases (or checks auto-assignment did it fairly) and rebalances when someone is on leave.
- 3. **Works the gate queue:** reviews packages prepared by executives, approves or sends back with comments.
- 4. **Decides escalations:** re-route after rejection, internal rejection, dormancy overrides, applicant complaints passed by Support.
- 5. **End of day:** checks ageing; leaves notes for the next day.

Permissions

Action	Allowed
View any application (masked PII by default)	Yes
Assign / reassign	Yes
Gate approve / send back	Yes, not on packages they prepared
Re-route decision, internal reject, close	Yes, with reason code
Edit application data	Only to correct with reason (logged)
Configure SLAs, holidays, assignment rules	Yes
Users, roles, permissions	No (Super Admin)
Bulk export	No (PAM request)

Feature map

Feature	What it does	Linked to
Control tower	Live view of queues and SLAs	Section 38

Feature	What it does	Linked to
Assignment console	Manual and rule-based assignment	Section 12
Gate queue	Packages waiting for approval	Section 17
Decision log	Re-route, reject, close with reason	Sections 21, 26
SLA settings	Timers per stage, working calendar	Jobs, notifications

Edge cases

- **Only one Ops Manager on duty and they prepared the package.** Gate approval falls to another Ops Manager or a Super Admin with approval rights; the system never allows self-approval.
- **Ops Manager leaves the company.** Super Admin deactivates the login; open approvals move to the queue.

29. Loan Ops Executive

In plain words: The Loan Ops Executive is the **case owner**. When an application lands in the back office and is assigned to them, it is their job from start to Ready for Bank: check the details, chase missing documents, pick the right bank from the applicant's consented list, and prepare the package. In Phase 2 they also follow the case while it sits with the bank and handle the bank's queries.

Figure 29 — Loan Ops Executive: sees, does, cannot

A day in the life

1. Opens **My queue**: new assignments, applicant replies, tasks due today, SLA timers.
2. Works each case in the **360° workspace** (section 13): details, documents, consent, timeline in one screen.
3. Raises **applicant tasks** for anything missing or unclear (plain-words message, secure link).
4. When everything is in order, runs **bank selection** (section 15) and builds the **submission package** (section 16), then sends it to the gate.

5. If sent back from the gate, fixes and resubmits. In Phase 2: records bank queries and follows up (section 20).

Permissions

Action	Allowed
View / edit own assigned cases	Yes
View other cases	No (Ops Manager can grant temporary cover)
Raise applicant tasks, add notes, call log	Yes
Accept / reject documents	Yes in MVP1 (Document Reviewer duty)
Select bank, build package	Yes
Approve at gate	Never on own package
Submit on bank portal	No (Operator)
Close application	Propose only; Ops Manager confirms

Feature map

Feature	What it does	Linked to
My queue	Personal work list with SLA timers	Sections 12, 38
360° workspace	Everything about one case on one screen	Section 13
Applicant tasks	Requests to the applicant	Sections 7, 20, 42
Bank selection tool	Picks one bank from consented list	Section 15
Package builder	Assembles and freezes the package	Section 16

Edge cases

- **Executive on leave.** Ops Manager reassigns; the timeline shows the handover.
- **Executive knows the applicant personally.** Must declare a conflict of interest; the case is reassigned.

30. Document Reviewer

In plain words: Banks reject many applications because of bad documents: blurry scans, expired IDs, salary slips that don't match the bank statement, or edited PDFs. The Document Reviewer looks at every file after the automatic security pipeline (section 8) has cleaned it, and decides: **accepted**, **rejected with a reason**, or **flag for fraud**. In MVP1 this is a duty performed by the Loan Ops Executive; when volume grows it becomes a separate role so the person checking documents is not the person selling the case.

Figure 30 — Document Reviewer: sees, does, cannot

What the reviewer checks

Check	Example
Readable and complete	All pages present, nothing cut off
Right document type	A salary slip, not an offer letter
Belongs to the applicant	Name and PAN match the profile (minor spelling differences noted)
Current	Statement covers the last 3 or 6 months the bank requires
Consistent	Salary on slip matches credits in bank statement
Not tampered	Pipeline hints: edited PDF metadata, font changes, mismatched totals

Rules

- Reviewers only ever see **watermarked previews** from the clean store; they never open the original upload or anything still in quarantine.
- Aadhaar images must show only the last 4 digits; if a full Aadhaar number is visible, the reviewer rejects and asks for a masked copy (we do not store full Aadhaar numbers).
- A rejection always has a reason code and a plain-words message to the applicant.
- Suspected tampering is never discussed with the applicant; the reviewer flags it and the case goes to Fraud & Risk (or the Ops Manager in MVP1).

Feature map

Feature	What it does	Linked to
Review screen	Preview, checklist, accept/reject	Sections 7–8, 14
Reason codes	Standard rejection reasons + messages	Communications (42)
Tamper hints	Shows pipeline signals	Section 8, Fraud (35)
Fraud flag	Hands case to Fraud & Risk	Section 35

Edge cases

- **Document in a regional language.** Accepted if the bank accepts it; reviewer notes the language.
- **Password-protected PDF.** The pipeline asks the applicant for the password once, unlocks it, and never stores the password.

31. Authorised DSA Operator

In plain words: Each bank gives its DSA a limited number of portal logins and wants to know exactly which named person uses each one. The Authorised DSA Operator is that person. They are the **only** people who can mark an application as submitted to a bank. Keeping this narrow protects the bank relationship: if something goes wrong on the portal, we know who did it and when.

Figure 31 — Authorised DSA Operator: sees, does, cannot

How someone becomes an operator

1. Bank/Partner Manager requests portal access from the bank for a named employee.
2. Bank issues credentials (stored in the company password vault, never in the platform).
3. Super Admin grants the Operator role and links the person to that bank in the **operator-bank register** with credential issue and expiry dates.
4. The operator sees only that bank's items in the submit queue.
5. When the person leaves or moves role, access is revoked the same day both in the platform and with the bank.

Permissions

Action	Allowed
View READY_TO_SUBMIT items for authorised banks	Yes
Download package files	Yes, via logged expiring links only
Record submission, bank ref, screenshot	Yes
Edit package or application data	No
Submit a package they prepared	No
Reply to bank queries on the portal	Yes, attaching what Ops prepared

Feature map

Feature	What it does	Linked to
Submit queue	Items per authorised bank	Section 18
Operator-bank register	Who may submit to which bank, expiry	Bank master (47), Super Admin
Submission record	Bank ref, screenshot, attestation	Attempt, audit
Download log	Every file fetched, by whom	Security (44)

Edge cases

- **Operator's credential expires mid-week.** Register alert 7 days before expiry (proposed); after expiry the queue hides that bank.
- **Bank wants one operator per branch.** The register supports bank + branch/hub level.

32. Bank/Partner Manager

In plain words: Every bank works differently: different products, document lists, portal screens, MIS formats, commission rates and contacts. The Bank/Partner Manager keeps all of that written down in the **bank master** (section 47) so the rest of the team and the

system follow the current rules. They are also the person who calls the bank’s area sales manager when cases are stuck.

Figure 32 — Bank/Partner Manager: sees, does, cannot

Main jobs

- 1. **Empanelment:** runs the paperwork to sign a DSA agreement with a new bank (section 47) and records its effective date.
- 2. **Bank master:** products, eligibility notes (as guidance only, not an engine), document checklists, serviceable pin codes, portal details, contacts, TAT expectations. Every change is versioned and approved by the Ops Manager.
- 3. **MIS:** receives MIS files, uploads them, maintains each bank’s column map and stage dictionary, clears unmatched rows with Ops.
- 4. **Commission schedules:** enters the rates from the agreement; Finance approves.
- 5. **Escalation:** monthly review with each bank; chases stale cases.
- 6. **Operator access:** requests and revokes portal logins with the bank.

Feature map

Feature	What it does	Linked to
Bank master editor	Versioned bank/product rules	Sections 15, 16, 46, 47
MIS console	Uploads, maps, exceptions	Section 19
Schedule drafts	Commission rules for Finance approval	Sections 24, 33
Bank scorecard	Approval rate, TAT, query rate per bank	Metrics (54)
Operator-bank register	Portal access per person	Section 31

Edge cases

- **Bank changes its document list overnight.** New bank-master version; packages already at the gate keep the version they were built with, and the Ops Manager decides whether to rebuild.

- **Bank agreement terminated.** Bank set to "inactive": no new attempts; open attempts continue to their outcome; consented lists keep the bank for history.

33. Finance

In plain words: Finance looks after the money side of the DSA business: what banks owe us, what they actually paid, tax deducted, invoices raised and disputes. Finance does not need to see applicants' documents and cannot, which keeps personal data exposure low.

Figure 33 — Finance: sees, does, cannot

Main jobs

1. Approve commission schedules drafted by the Bank/Partner Manager.
2. Upload payout files and remittances; run matching (section 25).
3. Work exceptions: disputes, adjustments (with a second approver), clawbacks.
4. Record TDS (section 194H) and reconcile with Form 26AS; manage GST invoices with the CA.
5. Month-end close: lock the period, publish the commission report.

Permissions

Action	Allowed
View ledger, payouts, loan account and amounts	Yes
View applicant name	Yes (needed to resolve disputes)
View documents, contact details	No
Approve schedules and others' adjustments	Yes
Approve own adjustment	No
Edit locked periods	No; corrections go in the next open period

Feature map

Feature	What it does	Linked to
Ledger views	Expected, actual, open items	Section 24
Payout matching	Classify every payout line	Section 25
Dispute tracker	Claims raised with banks	Section 32
Period lock	Closes a month	Audit, metrics (54)
Tax register	TDS, GST lines	Section 50

Edge cases

- **One Finance person only.** A second approver is still needed; the founder or another nominated approver holds the Finance-approver permission.

34. Compliance & Audit

In plain words: Compliance is the platform's conscience. They can look at everything that has happened — every consent, every status change, every download — but they cannot change applications. Their job is to prove to a bank, an auditor or a regulator that we did what we promised: we had consent, we shared data only with consented banks, we answered data requests on time, and we deleted data when we should.

Figure 34 — Compliance & Audit: sees, does, cannot

Main jobs

1. **Data-rights requests** (section 41): access, correction, erasure, grievance; tracked against legal timelines.
2. **Consent and notice text:** approves every new version before it goes live (with counsel).
3. **Audit reviews:** monthly sample of cases, downloads, gate approvals and PAM elevations; findings recorded.
4. **Retention and purge:** reviews purge logs; sets and lifts legal holds.

5. **Incident response:** with Super Admin, runs the security incident process, including CERT-In reporting within 6 hours where applicable (VERIFY).
6. **Compliance register** (section 51): keeps each obligation, owner and evidence up to date.

Permissions

Action	Allowed
Read audit log, consent ledger, purge log	Yes, all
Read applications	Yes, masked; unmasked via PAM with reason
Edit applications or documents	No
Approve consent/notice versions	Yes
Legal hold	Set and lift
Export audit reports	Yes, watermarked and logged

Feature map

Feature	What it does	Linked to
Audit explorer	Search by person, case, action, date	Section 39 audit event
Consent ledger view	Every consent version per applicant	Section 5
Rights request tracker	Requests with due dates	Section 41
Legal hold	Pauses purge	Section 26
Compliance register	Obligations and evidence	Section 51

Edge cases

- **Compliance needs to look at a document** for a complaint. PAM elevation, reason, time-limited, logged, and visible to the Super Admin.

35. Fraud & Risk

In plain words: A DSA's reputation with banks depends on not sending them fraud. Common patterns in Indian retail lending: edited salary slips and bank statements, borrowed identities, the same phone or address used by many "applicants", and agents coaching applicants to lie. Fraud & Risk watches for these signals and can place a **hold** on any application. They never accuse the applicant; the applicant only sees "We need more time to review your application".

Figure 35 — Fraud & Risk: sees, does, cannot

Signals (rules, not AI decisions)

Signal	Example	Source
Duplicate identity	Same PAN on two open applications	Registration, profile
Velocity	5 applications from one device in a day	Session data
Shared contact	One mobile or address across unrelated applicants	Profile
Document tamper hints	Edited PDF metadata, mismatched totals	Section 8 pipeline
Bank feedback	Bank rejects citing fraud or forged documents	Section 21
Staff behaviour	Unusual downloads or after-hours access	Audit log

Flow

1. A signal creates a **flag** on the case (with the evidence). High-severity flags place an automatic hold.
2. Fraud & Risk reviews and either clears the flag (with note) or opens an **investigation**.
3. Outcome: cleared → hold lifted; confirmed → close as REJECTED_INTERNAL, add identifiers to the internal deny list, block re-route, and inform the affected bank if a submission happened (as the DSA agreement requires).

4. If staff misconduct is involved, the case goes to the founder and HR.

Rules

- No automated rejection: every fraud decision is made by a person with a written note.
- Deny-list entries have a review date; personal data in them follows retention rules.
- In MVP1, the Ops Manager covers this role using the same flag screens.

Feature map

Feature	What it does	Linked to
Flag engine	Rule-based signals with evidence	Sections 4, 8, 44
Hold	Freezes the case	Status model (39)
Investigation record	Notes, evidence, outcome	Audit, Compliance (34)
Internal deny list	Blocks known bad identifiers	Registration (4)

Edge cases

- **Genuine family members share a phone.** Flag cleared with note; the shared-contact rule allows a whitelist per case.

36. Customer Support

In plain words: Applicants will call, WhatsApp or email to ask "What is happening with my loan?" Support answers using the same plain-words status the applicant sees in their tracker, logs complaints, and passes anything that needs action to the case owner. Support never sees documents, never changes data, and never promises approval.

Figure 36 — Customer Support: sees, does, cannot

Flow

1. **Identify** the caller: mobile OTP or reference number + date of birth. No data is shared until identity is confirmed.

2. **Answer** from the support view: current status, last update date, open applicant tasks, bank name (after submission).
3. **Act** within limits: resend a secure upload link, update contact preferences, record a withdrawal request.
4. **Hand off** anything else to the case owner as a note with a due date.
5. **Complaints** are logged as tickets with category and SLA, and escalated to the Grievance Officer if unresolved (section 41).

Rules

- Scripts are approved by Compliance: always say "Loans4India is a loan sourcing service, not a lender" when relevant, and never quote rates as ours.
- Any request for money in our name is treated as fraud and escalated.
- Calls may be recorded only with an announced notice (VERIFY).

Feature map

Feature	What it does	Linked to
Support view	Masked profile, status, history	Section 10 tracker
Ticketing	Complaints with SLA	Section 41 grievance
Link resend	New secure link, old one expires	Sections 7, 42
Hand-off note	Tasks for the case owner	Section 13 workspace

Edge cases

- **Caller is a relative** of the applicant. No details shared without the applicant's consent on record.
- **Applicant asks why a bank rejected them.** Support shares only what the bank allows and points to the bank's grievance channel.

37. Super Admin

In plain words: The Super Admin sets up the platform: who can log in, what role they have, and how the system is configured (SLA timers, templates, bank master publishing, feature switches). It is the most powerful role, so it is deliberately kept away from customer data

and casework, and every action it takes is audited and visible to Compliance.

Figure 37 — Super Admin: sees, does, cannot

Main jobs

- 1. **Joiner-mover-leaver:** create users with MFA, assign one role, disable on the day someone leaves.
- 2. **Role changes** need a second approver (another Super Admin or the founder).
- 3. **Configuration** is versioned: draft → approve → publish, with rollback.
- 4. **PAM approvals:** approves time-limited elevations for others (never for themselves).
- 5. **Access reviews:** monthly list of all users and roles, signed off with Compliance.
- 6. **System health:** job failures, storage, error rates; coordinates with engineering.

Rules

- At most two Super Admins (proposed). Break-glass account sealed and used only in an emergency, with alerting.
- Super Admin cannot view documents or unmasked PII without PAM elevation, and cannot approve their own elevation.
- Super Admin can act as backup gate approver only through a recorded PAM elevation.

Feature map

Feature	What it does	Linked to
User management	Create, disable, MFA reset	RBAC & PAM Spec
Role assignment	With second approver	Section 27
Config console	Versioned settings, templates	Sections 42, 46, 47
PAM console	Approve and audit elevations	Section 44
Access review	Monthly user/role sign-off	Compliance (34)

Edge cases

- **Only Super Admin is unavailable during an outage.** Break-glass procedure with the founder; all actions reviewed after.

Part 4 — How it fits together

38. Air-traffic control (the control tower)

In plain words: Like an airport tower that sees every plane, the **control tower** is one screen where managers see every application, where it is, how long it has been there, and who owns it. Without it, cases quietly get stuck. With it, nothing waits more than a day without someone noticing.

Figure 38 — Control tower: four queue groups in, three actions out

What it shows

Tile	Count	Drill-down
Intake	New today, unassigned, duplicates held	List → assign
In review	Per stage: verification, action needed, preparation	Per owner
Gate and submit	Waiting for approval, sent back, ready to submit	Per approver / operator
With bank	By bank status; queries due; stale (7+ days)	Per bank
Exceptions	Unmatched MIS rows, missing schedules, failed jobs	Per type
People	Cases per person, overdue per person, who is on leave	Per team

SLA colours

Every stage has a timer measured in **working hours** (holiday calendar applies). Green = within SLA; amber = 75% used; red = breached. Proposed SLAs: assignment 2 working hours; first review 1 working day; gate decision 1 working day; portal entry 2 working days after Ready for Bank. All configurable by the Ops Manager.

Actions from the tower

- **Alerts:** amber/red items notify the owner; red notifies the Ops Manager.
- **Rebalance:** bulk reassign, set cover for someone on leave, mark priority.
- **Daily report:** an automatic morning summary (funnel, ageing, exceptions) emailed to managers. No customer personal data in the email — only counts and case IDs.

Feature map

Feature	What it does	Linked to
Queue tiles	Live counts by stage	Status model (39)
SLA engine	Timers, colours, alerts	Job scheduler, calendar
Bulk actions	Reassign, prioritise	Assignment (12)
Daily report	Morning summary	Metrics (54)

Edge cases

- **Count looks wrong.** Every tile is a saved query on the same tables the workspaces use, so the tower and the lists can never disagree.

39. Single source of truth (data model and status model)

In plain words: "Single source of truth" means every fact lives in exactly one place in the system, and every screen, report and export reads from there. No side spreadsheets, no WhatsApp groups as records. If it isn't in the platform, it didn't happen. This section is the shared map for business and developers.

Figure 39 — Core entity model and the three status fields

Core entities

Entity	What it is	Key fields
Applicant	The person applying	id, mobile (verified), name, PAN (encrypted + hash for matching), DOB
Application	One loan request	id, reference (e.g. LFI-7K3P9Q), product, amount, state, owner, close reason
Consent record	One consent given or withdrawn	text shown, notice version, bank-list version, purpose, time, IP, device
Document	One uploaded file	type, pipeline status, review status, storage key, hash
Attempt	One submission to one bank	bank, state, bank reference, package hash, operator
Bank status event	One status update from a bank	status, source, evidence file, checker
Task / note	Work for a person	for applicant or staff, due date, status
Commission entry	Money expected or received	type, amount, schedule version, tax lines
Audit event	Anything that happened	actor, role, action, object, before/after, reason, time, IP

The three status fields

1. **Application state** (our process): ENQUIRY → CONSENT_PENDING → PROFILE_IN_PROGRESS → DOCUMENTS_PENDING → VERIFICATION_REVIEW ↔ ACTION_NEEDED → APPLICATION_PREPARATION → INTERNAL_REVIEW → READY_FOR_BANK → SUBMITTED_TO_BANK → CLOSED (+ reason). ON_HOLD can apply at any point.
2. **Attempt state** (one per bank): PREPARING → READY_TO_SUBMIT → SUBMITTED → OUTCOME_RECEIVED → CLOSED; or RETURNED_OR_FAILED; or SUPERSEDED on re-route.
3. **Bank-confirmed status** (from bank evidence only): RECEIVED, PROCESSING, QUERY, IN_PRINCIPLE, SANCTIONED, AGREEMENT_COMPLETED, DISBURSED, REJECTED, CANCELLED.

Keeping them separate stops a common mistake: mixing "what we did" with "what the bank said".

Rules for developers

- State changes happen only through named transitions in a state machine (e.g. `approveGate`, `recordSubmission`), each with a permission check and an audit event in the same database transaction.
- Consent records, bank status events, commission entries and audit events are **append-only**.
- IDs are random (UUID); applicant-facing references avoid confusing characters (no 0/O/1/I/L).
- Personal fields are encrypted at rest by tier (section 43); search uses hashes, not plain values.
- Every list and export reads from these tables; no separate reporting copy in MVP1.

Edge cases

- **Two people edit the same case.** Optimistic locking: the second save is rejected with "this case changed, reload".

40. Operating model

In plain words: The operating model explains how the business is organised around the platform: which layer does what, who owns each decision, and where the line sits between Loans4India and the bank. The short version: **applicants decide whether to apply; we decide how to prepare and where to send; the bank decides whether to lend.**

Figure 40 — Operating model: six layers and who decides in each

Who owns what (RACI)

R = does the work, A = accountable, C = consulted, I = informed.

Activity	Ops Exec	Ops Mgr	Operator	Bank Mgr	Finance	Compliance
Review and prepare	R	A	-	C	-	-

Activity	Ops Exec	Ops Mgr	Operator	Bank Mgr	Finance	Compliance
Bank selection	R	A	-	C	-	-
Gate approval	-	R/A	-	-	-	I
Portal entry	-	A	R	I	-	-
MIS and tracking	R	A	-	R	-	-
Commission schedules	-	-	-	R	A	-
Reconciliation	-	-	-	C	R/A	-
Consent and notices	-	C	-	-	-	R/A
Data-rights requests	C	C	-	-	-	R/A

Boundaries that never move

- We never lend, sanction, set rates or collect money from applicants.
- We never submit without consent covering that bank, or without a second-person gate approval.
- We never set a bank status without bank evidence.

Operating rhythm

- **Daily:** stand-up on the control tower; red items first.
- **Weekly:** bank review (stale cases, queries, rejection reasons).
- **Monthly:** commission close, access review, compliance sample audit, metrics review.

Part 5 — Platform foundations

41. Data rights and complaints (grievance)

In plain words: Under India's **DPDP Act 2023**, applicants ("data principals") can ask what data we hold, ask us to correct it, ask us to erase it, withdraw consent, and complain. As a DSA working for RBI-regulated banks, we must also run a **grievance** process with a named Grievance Officer whose contact details are published on the website. This section is how both work.

Figure 41 — Data-rights requests and the complaint ladder

Request types

Request	What we do	Target time (VERIFY)
Access	Summary of data held, purposes, banks shared with	Within statutory time; aim 7 days
Correction	Fix wrong data; inform the bank if already shared	7 days
Erasure	Purge unless law or an open bank process needs it; explain what remains	7 days to act; purge job runs
Withdraw consent	Close or stop processing (section 26)	Immediately
Nominate	Record a nominee for the applicant's data rights	7 days
Complaint	Level 1 Support → level 2 Grievance Officer	Acknowledge 1 day; resolve 15 days

Step by step

1. Request comes through the website form, email or phone; Support or Compliance logs it.
2. **Identity check:** OTP to the registered mobile. No data leaves until verified.
3. **Logged** with type, due date and owner; the applicant gets an acknowledgement with a ticket number.
4. **Data rights** go to Compliance; **complaints** start with Support.
5. Unresolved complaints escalate to the Grievance Officer; if still unresolved, the applicant is told their outside routes: the bank's grievance process (for bank decisions), RBI's Integrated Ombudsman via the bank, or the Data Protection Board for data matters.
6. Reply with evidence; ticket closed with outcome code.

Rules

- Grievance Officer name, email and phone shown on the website and in every applicant email footer.
- Complaints are never closed without the applicant being told the outcome.

- Repeated complaints about one bank or one staff member trigger a review.

Feature map

Feature	What it does	Linked to
Rights request form	Web form + OTP verification	Website, registration (4)
Request tracker	Type, due date, owner, evidence	Compliance (34)
Complaint ticketing	Levels, SLA, outcome codes	Support (36)
Data export	Applicant-readable summary	Data model (39)
Erasure action	Triggers purge with exceptions	Section 26

Edge cases

- **Erasure requested while the bank is still processing.** We stop our processing and erase what we can; we tell the applicant the bank holds its own copy and must be asked separately.

42. Communications (SMS, WhatsApp, email)

In plain words: The platform talks to applicants at key moments: OTP, application received, document needed, submitted to <Bank>, bank decision, disbursed. In India, business SMS must use **templates registered on the TRAI DLT** system (each with a template ID and an approved sender header), and WhatsApp business messages need pre-approved templates too. So messages are never typed freely by staff; they come from an approved library.

Figure 42 — Communications: template, checks, send, log

Message catalogue (core)

Trigger	Channel	Type
OTP	SMS	Transactional

Trigger	Channel	Type
Application received	SMS + email	Service
Document or answer needed	SMS + WhatsApp + email	Service
Reminder (task open)	SMS / WhatsApp	Service
Submitted to <Bank>	SMS + email	Service (disclosure)
Bank status change (key milestones)	SMS + email	Service
Disbursed	SMS + email	Service
Closed / withdrawn	Email	Service
Offers, newsletters	Email / WhatsApp	Marketing (separate consent)

Rules

- Templates are versioned and approved by Compliance; each stores DLT template ID, sender ID and language.
- Service messages go only for the application's purpose; marketing needs its own opt-in and has a one-tap opt-out.
- Quiet hours: no non-OTP messages outside 9am–9pm IST (proposed); held until morning.
- Messages never contain full PAN, account numbers or document images; links are short-lived and require OTP.
- All messages state "Loans4India is a loan sourcing service, not a lender" where space allows (email always).

Feature map

Feature	What it does	Linked to
Template library	Approved, versioned, DLT IDs	Compliance (34)
Rules check	Consent, opt-out, quiet hours	Consent ledger (5)

Feature	What it does	Linked to
Channel adapters	SMS, WhatsApp, email providers	Integrations
Delivery log	Sent, delivered, failed, retries	Audit, support (36)
Preference centre	Applicant manages channels	Applicant account

Edge cases

- **SMS fails.** Retry once; then fall back to email for service messages. OTP failures show "Resend" after 30 seconds.
- **Applicant replies to an SMS.** Short codes cannot receive replies; the message says how to reach us.

43. Data governance

In plain words: Data governance means knowing what data we have, why we have it, who can see it, where it lives and when it is deleted. We sort every field into one of four **tiers**; the tier decides the controls. The rule of thumb: **collect only what a bank actually needs, keep it only as long as needed, show it only to people who need it.**

Figure 43 — Data tiers T0-T3 with examples and controls

Data inventory (summary)

Data	Tier	Purpose	Kept until (proposed, VERIFY)
Mobile, name, email	T2	Contact, identity	Close + 180 days
PAN	T3	Identity, bank requirement	Close + 180 days
Address, employment, income	T2	Bank application	Close + 180 days
Identity and income documents	T3	Bank application	Close + 90 days (or bank agreement)

Data	Tier	Purpose	Kept until (proposed, VERIFY)
Consent records	T3	Proof of consent	8 years (VERIFY)
Audit events	T3	Proof of actions	8 years; personal fields minimised
Commission and tax records	T2	Accounting, tax law	8 years
Session and device logs	T2	Security, CERT-In	180 days (CERT-In)

Rules

- **Data minimisation:** each form field must name the bank requirement it serves; unused fields are removed.
- **Aadhaar:** we never store the Aadhaar number; images must be masked (last 4 digits only).
- **Location:** all personal data stored in India (cloud India region), including backups.
- **Access:** T2/T3 visible only within role scope; T3 previews watermarked; downloads logged.
- **Copies:** no personal data in emails, chat tools, spreadsheets or test environments. Test environments use synthetic data only.
- **Processors:** every vendor (SMS, storage, email) has a data processing agreement and an India-region setup where possible.

Feature map

Feature	What it does	Linked to
Field registry	Tier, purpose, retention per field	Data model (39)
Masking	Shows partial values by role	RBAC (27)
Encryption	Field-level for T3, disk for all	Security (44)
Retention table + purge	Deletes on schedule	Section 26
Vendor register	Processors and agreements	Compliance register (51)

44. Security

In plain words: We hold exactly the data fraudsters want: identity documents, bank statements and phone numbers of people who need money. Security is therefore built in layers, so that if one fails the next still protects the data. Banks will also audit us before giving portal access, so these controls are part of getting empanelled.

Figure 44 — Security: five layers of defence

Controls by layer

Layer	Controls	Owner
Edge	TLS 1.2+, WAF, rate limits on OTP and uploads, bot checks, DDoS protection	Engineering
Identity	Applicant OTP with retry limits; staff MFA, SSO later, 15-minute idle timeout (proposed), managed devices for T3 access	Super Admin
Application	Permission + object-level check on every API (prevents seeing another person's case by changing an ID), validation, CSRF protection, signed short-lived URLs	Engineering
Data	Encryption at rest; T3 fields encrypted with KMS keys; quarantine and clean stores separated; India-region storage and backups; daily backups with restore tests	Engineering
Watch and respond	Audit log, anomaly alerts (bulk views, after-hours access), logs kept 180 days, incident runbook, CERT-In reporting within 6 hours (VERIFY scope)	Compliance + Super Admin

Secure development

- Code review for every change; dependency scanning; secrets never in code.
- Security tests before go-live: access-control tests for every role, file-upload attacks, OTP brute force, and an external penetration test before live data (Phase 2 gate).

- Separate environments; test environments never hold real personal data.

Incident response (summary)

1. Detect (alert or report) → 2. Contain (disable accounts, rotate keys) → 3. Assess what data was affected → 4. Notify: CERT-In (within 6 hours, where applicable), affected banks per agreement, Data Protection Board and affected people as DPDP requires (VERIFY) → 5. Fix and review.

Feature map

Feature	What it does	Linked to
Permission middleware	Checks role and scope on every call	RBAC (27)
Upload pipeline	Quarantine, scan, clean store	Section 8
KMS integration	Field-level encryption keys	Data governance (43)
Anomaly alerts	Unusual access patterns	Compliance (34), Fraud (35)
Incident runbook	Steps, contacts, templates	Compliance register (51)

45. Affiliates and referral partners

In plain words: Many loan applicants come through people they already trust: chartered accountants, property brokers, car dealers, employers’ HR teams. These **referral partners** can send applicants to Loans4India using a tracked link or code and earn a share of our commission. The key design rule: **partners send people, not data**. The applicant always registers, consents and uploads with us directly, so we never receive personal data collected by someone else without proper consent.

Figure 45 — Affiliates and referral partners: link in, payout out

Step by step

1. **Onboard the partner:** agreement (scope, conduct, payout share), partner KYC (PAN, GST if applicable, bank account), unique code. Approved by the Ops Manager.
2. **Share link:** partners use approved wording only ("Apply through Loans4India, a loan sourcing service"). No rate promises, no bank logos unless permitted.
3. **Applicant applies** with their own OTP and consent. The consent screen does **not** share data back with the partner.
4. **Source tagged:** the application records the partner code (and UTM details) for attribution.
5. **Payout:** after the bank pays us for that disbursal (section 25), the partner's share is calculated, TDS deducted, and paid monthly.
6. **Statement:** partners see counts and payout amounts by month, and at most an application reference — no names, statuses or documents unless the applicant opts in.

Rules

- A partner who collects documents themselves and uploads on behalf of applicants is **not allowed** in MVP1/Phase 2 (VERIFY with counsel before ever allowing).
- Partners may not charge applicants any fee. Breach ends the agreement.
- Partner payouts are only from commission actually received.

Feature map

Feature	What it does	Linked to
Partner register	Agreement, KYC, code, status	Bank/Partner Manager (32)
Tracked links	Attribution to partner	Entry points (section 1)
Partner payout	Share of received commission, TDS	Finance (33), ledger (24)
Partner statement	Counts and payouts only	Data governance (43)

Edge cases

- **Two partners claim the same applicant.** First valid tracked click within 30 days wins (proposed).

- **Applicant asks us not to tell the partner anything.** Default already: statements carry no personal data.

46. Product catalogue

In plain words: The catalogue lists the loan types we source (for example personal loan, business loan) and, for each partner bank, that bank’s version of the product: what documents it wants, where it lends, what notes staff should know, and what it pays us. It is **not** an eligibility engine: it does not decide who qualifies. It helps staff pick the right bank and build the right package.

Figure 46 — Product catalogue structure

Products by phase (proposed)

Product	Phase	Notes
Personal loan (salaried)	MVP1	Synthetic data in MVP1
Business loan (self-employed)	MVP1	Synthetic data in MVP1
Home loan, balance transfer	Phase 3	Property documents, longer timelines, tranches
Loan against property	Phase 3	Valuation and legal checks by the bank
Car / two-wheeler loan	Later	Dealer partners (section 45)

Bank product fields

Field	Example	Used by
Applicant types	Salaried, self-employed professional	Application form (6)
Document checklist	3 months' slips, 6 months' statement, PAN, address proof	Document center (7), package (16)
Guidance notes	"Prefers employers in category A/B"	Bank selection (15)
Service areas	Pin codes / cities	Bank selection (15)

Field	Example	Used by
Amount and tenure range	₹50,000–₹25 lakh, 12–60 months	Quick estimate (3), selection
Portal field map	Our field → bank portal field	Portal entry (18)
Commission schedule	Link to versioned schedule	Commission (24)

Rules

- Every change is a new version with effective dates; the Ops Manager approves before publish.
- Packages remember the version they were built with.
- Public website shows only T0 content; partner bank names appear only once an agreement is live.

47. Bank empanelment (adding a partner bank)

In plain words: Empanelment is the process of becoming an approved DSA of a bank. There is no RBI "DSA licence"; each bank or NBFC signs its own DSA agreement after checking us. As of 2 Oct 2026 no agreement is signed, so the platform is built to switch a bank on only when every piece is ready.

Figure 47 — Bank empanelment: eight steps to go live

Go-live checklist (per bank)

Item	Owner	Evidence
Signed DSA agreement + effective date	Founder / Bank Manager	PDF in contract store
Bank due diligence passed (company KYC, security questionnaire)	Founder	Bank confirmation

Item	Owner	Evidence
Staff registered with the bank, code of conduct signed	Bank Manager	Register
Portal logins issued to named operators	Bank Manager	Operator-bank register
MIS format and payout format received; column maps built	Bank Manager	Map versions
Bank master: products, documents, service areas, portal field map	Bank Manager, Ops Manager approves	Published version
Commission schedule entered and approved	Bank Manager, Finance approves	Schedule version
Partner-bank list and consent text updated, approved by Compliance	Compliance	New notice + bank-list version
Website disclosure updated (partner list, grievance)	Compliance	Page version

Rules

- The bank stays **inactive** in the platform until every item has evidence; the system blocks attempts to inactive banks.
- Adding a bank creates a new bank-list version. Existing applicants are not assumed to consent to it; they need one-tap re-consent if we ever want to send to the new bank (section 5).
- Ending an agreement: bank set inactive, open attempts finish, bank removed from new consent versions.

Feature map

Feature	What it does	Linked to
Bank record + status	Inactive / live / suspended / ended	Bank master, section 15
Go-live checklist	Evidence per item, blocks activation	Compliance (34)
Bank-list versioning	Feeds consent screen	Section 5

48. AI and automation policy

In plain words: Software and AI can save staff time, but in lending they can also cause unfair or unexplainable outcomes. Our rule is simple: **automation can help a person; it never decides about a person.** Every AI feature is listed, tested and switchable off.

Use	Allowed?	Conditions
Document type detection and quality hints (blur, cropped)	Yes	Shown as a hint; reviewer decides
Reading fields from documents (OCR) to pre-fill	Yes	Person confirms every value
Tamper / duplicate signals	Yes	Creates a flag for a person (section 35)
Summarising case notes for staff	Yes, Phase 2	No new facts; source notes linked
Choosing the bank automatically	No	Staff choose; tool may rank by stated rules only
Approving, rejecting or holding an applicant	No	Always a named person
Chatbot giving loan or rate advice	No	FAQ bot with approved answers only, Phase 3
Robotic entry into bank portals (RPA)	No	Only with a bank's written approval
Sending personal data to an outside AI service	No by default	Only with a data processing agreement, India hosting where required, and Compliance approval

Controls: an AI register (feature, model, vendor, data used, owner, last test); accuracy checks on synthetic and sampled data; every AI output stored with the human decision next to it; a kill switch per feature.

49. Non-functional requirements (NFRs)

In plain words: Functional requirements say *what* the platform does; non-functional requirements say *how well*: how fast, how reliable, how secure, how accessible. These are targets the development team designs and tests against (all proposed; confirm before build).

Area	Target (MVP1 → Phase 2)
Speed (applicant)	Pages usable in under 3 s on a mid-range phone on 4G; API p95 under 500 ms
Speed (staff)	Workspace opens under 2 s; lists of 1,000 cases paginate under 1 s
Uploads	Up to 10 MB per file, JPG/PNG/PDF; resumable on weak networks; scan result within 60 s
Availability	99.5% monthly (MVP1) → 99.9% (Phase 2), excluding announced maintenance
Capacity	5,000 applications/month at launch; design for 50,000
Backups	Daily (MVP1), point-in-time (Phase 2); restore tested quarterly; RPO 24 h → 1 h; RTO 8 h → 4 h
Data location	All personal data and backups in an India region
Security	Section 44 controls; external penetration test before live data
Accessibility	WCAG 2.1 AA for applicant pages; large tap targets; works without perfect literacy (icons + simple words)
Languages	English at launch; Hindi in Phase 2; text kept in translation files from day one
Devices	Mobile-first; latest two versions of Chrome, Safari, Edge; Android 9+
Observability	Error tracking, uptime checks, job dashboards, alert within 5 minutes of failure

Area	Target (MVP1 → Phase 2)
Audit	100% of state changes, views of T3 data and downloads produce audit events
Logs	Security logs kept 180 days (CERT-In) in India

50. Legal and regulatory framework

In plain words: This is the list of laws and rules that shape the platform. Loans4India is not a lender, so most banking rules apply to the banks we work for, and reach us through our DSA agreements. Some rules apply to us directly (data protection, IT security, telecom messaging, tax). Each item needs counsel confirmation (VERIFY) before live data.

Law / rule	What it means for us	Where in this doc
RBI Digital Lending Directions, 2025 (for banks and their LSPs)	If we are a Lending Service Provider for a bank's digital journey: list partner banks on our site, grievance officer, no fees from borrowers, data stored in India, KFS by the bank. VERIFY our classification	Sections 5, 22, 41, 43
RBI rules on banks' outsourcing and DSA conduct (Fair Practices Code, DSA code of conduct)	Banks pass these to us via the agreement: identify ourselves, no misleading claims, no harassment, staff training	Sections 27–37, 42
DPDP Act 2023 and DPDP Rules 2025 (phased commencement, VERIFY dates)	Consent notices, purpose limits, data-principal rights, breach notice, retention limits	Sections 5, 26, 41, 43
IT Act 2000 and CERT-In Directions 2022	Reasonable security, incident reporting within 6 hours, logs kept 180 days in India	Section 44
TRAI TCCCPR 2018 (DLT)	Registered headers and templates for SMS; consent for promotional messages	Section 42

Law / rule	What it means for us	Where in this doc
Aadhaar Act and UIDAI rules	No Aadhaar authentication by us; never store the number; masked images only	Sections 7, 43
Income Tax Act section 194H	Banks deduct TDS on our commission; we deduct TDS on partner payouts	Sections 25, 45
GST law	GST on commission invoices to banks (VERIFY treatment with CA)	Section 25
Consumer protection law	Fair advertising; no false promises of approval or rates	Website copy, section 42
Contract: each DSA agreement	Data handling, retention, audits, conduct, payout terms	Sections 47, 51

Documents we must publish or hold: privacy notice, terms of use, grievance page, partner-bank list, consent texts (versioned), DSA agreements, vendor data processing agreements, incident runbook.

51. Compliance register

In plain words: One table that lists every obligation, how the platform meets it, who owns it and what proves it. Compliance reviews it monthly; banks and auditors can be shown it. Status values: **Designed** (in this doc), **Built**, **Verified** (counsel or auditor confirmed).

#	Obligation	Source	Platform control	Owner	Evidence	Status
1	Informed, specific consent naming partner banks	DPDP; DSA agreements	One consent screen, versioned text and bank list (s.5)	Compliance	Consent ledger	Designed, VERIFY
2	Tell applicant which bank received the application	Lender identification; fair practice	SMS/email at submission (s.18)	Ops Manager	Delivery log	Designed, VERIFY

#	Obligation	Source	Platform control	Owner	Evidence	Status
3	Share only with consented banks	DPDP purpose limit	Consent re-check at selection and portal entry (s.15, 18)	Ops Manager	Audit events	Designed
4	No fees from applicants	RBI digital lending; DSA code	No payment features; support scripts; fraud alerts (s.22)	Founder	Policy, complaints log	Designed
5	Grievance officer published and complaints handled	RBI; DPDP	Grievance page, ticketing, SLAs (s.41)	Compliance	Ticket reports	Designed
6	Data-principal rights	DPDP	Rights tracker, export, erasure (s.41)	Compliance	Tracker	Designed
7	Retention limits and erasure	DPDP; bank agreements	Retention table, purge job, purge log (s.26)	Compliance	Purge log	Designed, VERIFY periods
8	Data stored in India	RBI digital lending; bank agreements	India-region hosting and backups (s.43)	Engineering	Cloud config	Designed
9	Reasonable security + incident reporting 6 h	IT Act; CERT-In	Section 44 controls, runbook	Super Admin	Pen test, drills	Designed
10	Logs kept 180 days	CERT-In	Log retention setting (s.44)	Engineering	Config	Designed
11	Registered SMS templates and headers	TRAI DLT	Template library with DLT IDs (s.42)	Compliance	DLT portal records	Designed
12	No Aadhaar number storage	Aadhaar Act	Masked images only, reject unmasked (s.30)	Compliance	Review samples	Designed
13	Bank status only from bank evidence	DSA agreements; accuracy	Evidence-required status events (s.19)	Bank Manager	Event evidence	Designed

#	Obligation	Source	Platform control	Owner	Evidence	Status
14	Separation of duties on submission and money	Internal control; bank audits	System-enforced SoD (s.17, 25, 27)	Super Admin	Audit reports	Designed
15	TDS and GST on commission	Income Tax s.194H; GST	Tax lines and register (s.25)	Finance	Form 26AS match	Designed, VERIFY with CA
16	Staff conduct and training	DSA code of conduct	Training record before role grant (s.37)	Super Admin	Training log	Designed

52. MVP1 scope

In plain words: MVP1 is the first version we build. Because no bank agreement is signed yet, MVP1 uses **synthetic (made-up) data only** and stops at **Ready for Bank**. That proves the hard parts — consent, documents, review, the gate, roles, audit — before any real customer data or bank is involved.

Figure 52 — MVP1 scope: everything up to Ready for Bank

In scope

Area	MVP1 content
Applicant journey	Entry pages, quick estimate, OTP registration, one consent screen, application details, document upload with full security pipeline, submit, status tracker (sections 1–10)
Back office	Intake, assignment, 360° workspace, review, applicant tasks, bank selection from one configurable bank template, package, Ready-for-Bank gate (sections 11–17)
Roles	Super Admin, Ops Manager, Loan Ops Executive (incl. document review), Compliance & Audit (proposed)

Area	MVP1 content
Controls	RBAC + PAM, SoD at gate, audit log, consent ledger, data tiers, retention table (purge job runs on synthetic data)
Ops tools	Control tower (basic), SLA timers, CSV export of counts only

Out of scope (Phase 2+)

Portal entry, MIS tracking, bank queries, disbursal, commission, reconciliation, partners, Hindi, AI features, live data.

Definition of done

- Every flow in sections 1–17 works end to end with synthetic applicants.
- Every role's *Cannot* list is proven by automated tests.
- Every state change creates an audit event; consent re-check blocks off-list banks.
- Security checks of section 44 pass; restore from backup demonstrated.
- Business sign-off by JP on a walkthrough of 10 synthetic cases.

53. Roadmap

In plain words: We grow in phases. Each phase starts only when its **gate** is met, because the next phase depends on something outside our control (a bank agreement, a bank approving an API). Dates are set once the gate is in sight.

Figure 53 — Roadmap: four phases and their gates (not to scale)

Phase	Starts when	Main additions
MVP1 Foundation	Now	Sections 1–17 on synthetic data; 4 roles

Phase	Starts when	Main additions
Phase 2 Bank operations	First bank or master-DSA agreement signed, penetration test passed, counsel sign-off on consent	Live data, Operator, Bank Manager, Finance, Support and Fraud roles; portal entry, MIS, queries, disbursement, commission, reconciliation
Phase 3 Automation	A bank approves API or bulk upload in writing	API/bulk handoff per bank, more products, Hindi, referral partners, fraud tooling, AI hints
Phase 4 Scale	Three or more banks live and volume needs it	Multi-bank operations, partner portal, analytics, split services if needed

54. Metrics

In plain words: Metrics tell us whether the platform is working for applicants, staff, banks and the business. Each one has a single definition so everyone counts the same way. All come from the event data in section 39.

Group	Metric	Definition
Funnel	Start → consent rate	Consented applications ÷ OTP-verified starts
Funnel	Completion rate	Submitted by applicant ÷ consented
Funnel	Ready-for-Bank rate	Reached READY_FOR_BANK ÷ submitted by applicant
Funnel	Sanction rate	SANCTIONED ÷ submitted to bank (per bank)
Funnel	Disbursement rate	DISBURSED ÷ submitted to bank (per bank)
Speed	Time to first review	Applicant submit → first staff action (working hours)
Speed	Time to Ready for Bank	Applicant submit → READY_FOR_BANK
Speed	Portal lag	READY_FOR_BANK → SUBMITTED_TO_BANK

Group	Metric	Definition
Speed	Bank turnaround	SUBMITTED → bank decision (per bank)
Quality	Gate send-back rate	Sent back ÷ gate decisions
Quality	Bank query rate	Attempts with ≥1 query ÷ submitted
Quality	Rejection reasons	Share of each reason code (per bank)
Quality	Document rejection rate	Rejected files ÷ reviewed files
Money	Commission realisation	Received ÷ expected (per bank, per month)
Money	Days to payout	Disbursal date → payout received
Money	Open disputes	Count and value by age bucket
Compliance	Rights requests on time	Closed within target ÷ total
Compliance	Purge on time	Purged by due date ÷ due
Compliance	Complaints per 1,000 applications	Complaints ÷ applications × 1,000
People	SLA breaches per person	Red items per owner per week

55. Open decisions and change log

Open decisions

#	Decision	Proposed	Owner
1	First bank or master DSA, and its channel	Start as sub-DSA under a master DSA if faster	JP
2	One-screen list consent without a per-submission notice meets DPDP and bank rules	Yes, with automatic SMS at submission naming the bank	Counsel
3	Are we a Lending Service Provider for any bank's digital journey?	Assume yes for website disclosures	Counsel

#	Decision	Proposed	Owner
4	Retention period per data class	Table in section 43	Counsel + Compliance
5	Maximum re-routes	2 (so at most 3 banks)	JP
6	Dormancy timers	7 / 30 / 15 days (section 26)	JP
7	Fourth MVP1 role	Compliance & Audit	JP
8	SLA values	Section 38	Ops Manager
9	PAN / name-match provider	Choose in Phase 2	Engineering
10	Background job runner	pg-boss on Postgres	Engineering
11	Named Grievance Officer	To appoint before live data	JP
12	Referral partner programme start	Phase 3	JP

Change log

Version	Date	Change
v1	Before Oct 2026	Marketplace / lender-matching model. Retired.
v2	3 Oct 2026	Virtual Corporate DSA model; lifecycle; guardrails.
v3 / v3.1	3 Oct 2026	All verticals; pre-submission notice and objection window.
v3.2	3 Oct 2026	Applicant no longer chooses the bank; notice/objection removed; one consent screen naming banks.
v4	3 Oct 2026	Rebuilt from the old heading list. Removed: eligibility engine, matched offers, credit check. Renamed: quick estimate, bank selection, Ready-for-Bank gate, bank portal entry. Added: Authorised DSA Operator, control tower, single source of truth, operating model, empanelment, affiliates, AI policy, compliance register. A diagram for every flow and every role.